Computer Methods in Applied Mechanics and Engineering 452 (2026) 118731

Contents lists available at ScienceDirect

Comput. Methods Appl. Mech. Engrg.

journal homepage: www.elsevier.com/locate/cma

A vertex-centered finite volume method for solute transport in porous media on arbitrary polygonal meshes

Yingzhi Qian [a], Xiaoping Zhang [b,*], Yan Zhu [a], Lili Ju [c], Jiesheng Huang [a]

[a] State Key Laboratory of Water Resources Engineering and Management, Wuhan University, Wuhan, 430072, Hubei, China
[b] School of Mathematics and Statistics, Wuhan University, Wuhan, 430072, Hubei, China
[c] Department of Mathematics, University of South Carolina, Columbia, SC, 29208, USA

ARTICLE INFO

2010 MSC:
65M08
65M50

Keywords:
Solute transport
Advection-diffusion equation
Sharp solute gradient
Vertex centered finite volume method
Nonmatching grid

ABSTRACT

Accurately modeling solute transport in porous media is essential for effective agricultural water resource management. However, strong advection or localized sink/source terms often leads to steep local concentration gradients, posing significant challenges for numerical simulation. To address these issues, this paper proposes an efficient algorithm based on the Vertex-Centered Finite Volume Method (VCFVM). In this approach, the primary unknowns are defined at the mesh vertices, and the fluxes across dual edges are computed as weighted combinations of these vertex values. This formulation enables the advection-diffusion equation to be directly solved on arbitrary polygonal meshes-including nonmatching grids-without requiring complex preprocessing. The use of nonmatching grids allows for flexible local refinement, significantly enhancing the method's ability to capture sharp concentration gradients. The proposed algorithm is particularly well-suited for simulating solute transport characterized by sharp gradients induced by strong advection or localized sink/source terms, and it has been validated through three representative benchmark cases. Numerical results show that the method accurately reproduces solute transport behavior in complex geometries, effectively maintains local Péclet numbers below 4, and successfully reduces numerical oscillations while suppressing excessive diffusion, as well as save about 50% CPU time in field scale simulation. This algorithm can be seamlessly integrated with existing groundwater and soil water flow models, offering a practical solution for managing sharp gradients in both water flow and solute transport.

1. Introduction

Accurate modeling of solute concentration in groundwater and soil is essential for sustaining agricultural productivity and environmental health [1]. Agricultural practices such as irrigation [2], flood leaching [3], and surface/subsurface drainage [4] are often implemented to regulate solute distribution, thereby preventing soil salinization and groundwater contamination. To support decision-making and optimize these interventions, robust numerical models are needed to simulate solute transport under various field conditions.

Among existing models, those based on the Advection-Diffusion Equation (ADE) provide a physically sound foundation for simulating solute transport. Tools such as HYDRUS [5] and MT3D [6] have been widely used in both soil and groundwater contexts.

* Corresponding author.
E-mail addresses: yzqian@whu.edu.cn (Y. Qian), xpzhang.math@whu.edu.cn (X. Zhang), zyan0701@163.com (Y. Zhu), ju@math.sc.edu (L. Ju), sdjshuang@whu.edu.cn (J. Huang).

https://doi.org/10.1016/j.cma.2026.118731
Received 10 August 2025; Received in revised form 29 November 2025; Accepted 5 January 2026
Available online 14 January 2026
0045-7825/© 2026 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

These models have been validated across diverse spatial and temporal scales [7–10]. However, their performance often degrades when simulating scenarios with sharp solute gradients, such as those caused by localized sources or advection-dominated flow fields.

Accurately resolving steep solute gradients requires local mesh refinement to avoid spurious oscillations and excessive numerical diffusion. Because these high-gradient regions typically occupy only a small portion of the domain, uniformly fine grids are computationally inefficient. This gives rise to multiscale challenges, where locally refined grids-with fine grids near critical features and coarse grids elsewhere-can offer a balance between accuracy and efficiency [11,12]. In certain contexts, upscaling methods [13] or flux-limiting schemes [14] may be used to reduce computational costs, but they are less suited for applications that demand local resolution of solute dynamics. Therefore, this study focuses on numerical methods that support local refinement for solving the ADE.

Several families of methods have been developed to support local mesh refinement. Traditional approaches such as finite element (FEM) [5,15], finite volume (FVM) [16,17], and discontinuous Galerkin methods (DGM) [18] enable local mesh refinement via triangular or polygonal elements. However, these often require redundant transitional meshes to avoid distortion. Domain Decomposition Methods (DDMs) provide an alternative by partitioning the domain into subdomains with different resolutions and solving them separately, using boundary conditions to couple the interfaces [19–21]. Mortar methods improve this interface coupling by introducing intermediate function spaces [22,23]. Meanwhile, multiscale methods such as the Multiscale Finite Element Method (MsFEM) [24] and Multiscale Finite Volume Method (MsFVM) [25] reconstruct basis functions or fluxes to capture submesh effects. These methods have been extended and refined in recent years [26–32].

Despite these developments, several limitations persist: (i) many methods require complex treatments at nonmatching mesh interfaces, such as interpolation or flux projection, increasing algorithmic overhead; (ii) under advection-dominated conditions or strong source/sink terms, they may exhibit excessive numerical dissipation or instability; and (iii) some methods are less adaptable to unstructured polygonal meshes commonly encountered in real-world hydrogeologic settings. To address these challenges, we propose an efficient and flexible numerical scheme based on the Vertex-Centered Finite Volume Method (VCFVM). The core idea is to define primary unknowns at mesh vertices and construct a dual control volume around each vertex. This structure eliminates the need for special interface treatments, naturally supports nonmatching grid refinement, and enables compact and consistent flux formulations. As a result, the proposed method achieves high accuracy and numerical stability, even under sharp-gradient or multiscale conditions, and is particularly well suited to complex agricultural and environmental modeling scenarios.

The rest of this paper is organized as follows: Section 2 introduces the governing ADE and describes the spatial-temporal discretization scheme. Section 3 outlines the design of the test cases. Section 4 presents the numerical results and discusses the method's performance in terms of accuracy, efficiency, and adaptability. Section 5 concludes the paper with final remarks and future research directions.

2. Solute transport problem and its full discretization using VCFVM

2.1. Governing equations

Let $\Omega \subset \mathbb{R}^2$ be a two-dimensional domain filled with porous media. Denote the initial spatial domain as $\Omega^0 = \Omega \times \{0\}$, and the full spatio-temporal domain as $\Omega^T = \Omega \times (0, T]$. The Dirichlet and Neumann boundaries extended over time are denoted by $\Gamma_D^T = \Gamma_D \times (0, T]$ and $\Gamma_N^T = \Gamma_N \times (0, T]$, respectively. The solute transport within this domain is governed by the Advection-Diffusion Equation (ADE):

$$\nabla \cdot (\theta \boldsymbol{D} \nabla c - \boldsymbol{q} c) = \frac{\partial \theta c}{\partial t} + f(\boldsymbol{x}, t), \quad (\boldsymbol{x}, t) \in \Omega^T, \tag{1a}$$

$$c(\boldsymbol{x}, t) = g_0(\boldsymbol{x}), \quad (\boldsymbol{x}, t) \in \Omega^0, \tag{1b}$$

$$c(\boldsymbol{x}, t) = g_D(\boldsymbol{x}), \quad (\boldsymbol{x}, t) \in \Gamma_D^T, \tag{1c}$$

$$(\theta \boldsymbol{D} \nabla c - \boldsymbol{q} c) \cdot \boldsymbol{n}(\boldsymbol{x}) = g_N(\boldsymbol{x}), \quad (\boldsymbol{x}, t) \in \Gamma_N^T, \tag{1d}$$

where c denotes the solute concentration to be determined, with units $[\mathrm{ML}^{-2}]$; θ is the volumetric water content, which remains constant in a homogeneous aquifer but varies spatially in soil, and is dimensionless $[-]$; $\boldsymbol{q}$ is the Darcy velocity vector in the porous medium, with components q_x and q_y in the x- and y-directions, respectively, measured in $[\mathrm{LT}^{-1}]$; g_0, g_D, and g_N represent the prescribed values for the initial condition, Dirichlet boundary condition, and Neumann boundary condition, respectively; $\boldsymbol{n}$ is the unit outward normal vector on the Neumann boundary $\partial\Omega$; f denotes the source or sink term, with units $[\mathrm{ML}^{-2}\mathrm{T}^{-1}]$, and will be discussed in detail in Section 3.4; $\boldsymbol{D}$ is the hydrodynamic dispersion tensor, with components D_{ij}, and units $[\mathrm{L}^2\mathrm{T}^{-1}]$, defined as follows:

$$D_{ij} = \alpha_T |\boldsymbol{v}| \delta_{ij} + (\alpha_L - \alpha_T) \frac{v_i v_j}{|\boldsymbol{v}|}, \tag{2}$$

where α_L and α_T are the longitudinal and transverse dispersivities, respectively, with units [L]; and δ_{ij} is the Kronecker delta symbol; $\boldsymbol{v}$ is the pore water velocity vector, and $|\boldsymbol{v}|$ is its magnitude with unit $[\mathrm{LT}^{-1}]$. Here, the soil is assumed to be homogeneous porous media within representative element volume, and is variably saturated, therefore, the relation between $|\boldsymbol{v}|$ and $|\boldsymbol{q}|$ could be expressed as: $|\boldsymbol{v}| = |\boldsymbol{q}|/\theta$.

In Eqs. (1) and (2), two variables related to the water flow, i.e., θ and $\boldsymbol{q}$, are explicitly or implicitly incorporated. Here, the solute transport is assumed to occur in variable saturated porous media, either two-dimensional soil profiles or unconfined aquifers. The Darcy flow $\boldsymbol{q}$ is defined as:

$$\boldsymbol{q} = \boldsymbol{K} \nabla H, \tag{3}$$

where $\boldsymbol{K}$ is the hydraulic conductivity tensor; H is the water total potential, [L]. It represents groundwater level for groundwater flow and is defined as follow for soil water problem:

$$H = h + z, \tag{4}$$

where h and z are soil matrix potential and vertical coordinate, respectively. For both unconfined aquifer and soil profile, H is a spatial-temporal variable.

Without considering buoyancy term, gravity and viscosity of the air phase, and the water is assumed to be impressible [33]. The water potential for groundwater and soil water can obtained by solving Groundwater Flow Equation (GFE) and mixed-form Richards' Equation (mRE) via VCFVM [34,35], respectively. The GFE is expressed as:

$$\nabla \cdot (\boldsymbol{K} H \nabla H) = \mu_d \frac{\partial H}{\partial t} + f_{gw}(\boldsymbol{x}, t), \quad (\boldsymbol{x}, t) \in \Omega^T, \tag{5}$$

where μ_d is specific yield of the unconfined aquifer, [-]; f_{gw} is sink/source term of groundwater flow.

The mRE is expressed as follow:

$$\nabla \cdot (\boldsymbol{K}(h) \nabla (h + z)) = \frac{\partial \theta(h)}{\partial t} + f_{sw}(\boldsymbol{x}, t), \quad (\boldsymbol{x}, t) \in \Omega^T, \tag{6}$$

where $f_{sw}(\boldsymbol{x}, t)$ is sink/source term of soil water flow, with unit [L^2T^{-1}].

It should be noted that in Eq. (5), the porous media is fully saturated, with θ being a time-independent term with value of θ_s. Therefore, H is the sole unknown in Eq. (5). However, more than one unknowns is introduced in Eq. (6), and constitutive relations describing $\theta(h)$ and $K(h)$ are required to close the equation system. In the VCFVM, the widely used van Genuchten model [36] is adopted, which is defined as:

$$\theta(h) = \begin{cases} \theta_r + \dfrac{\theta_s - \theta_r}{(1 + |\alpha h|^n)^m}, & h < 0, \\ \theta_s, & h \geq 0. \end{cases} \tag{7}$$

where α is a parameter related to the mean pore-size, n is a parameter related to the pore-size distribution, and $m = 1 - 1/n$. The quantitative relation between $\theta(h)$ and h is defined as:

$$K(h) = K_s S_e^{0.5} \left(1 - (1 - S_e^{\frac{1}{m}}) \right)^2, \tag{8}$$

where K_s is the saturated hydraulic conductivity and S_e represents the effective saturation that defined as:

$$S_e = \frac{\theta(h) - \theta_r}{\theta_s - \theta_r}. \tag{9}$$

Within each time step, Eq. (5) or Eq. (6) is solved firstly, then θ and $\boldsymbol{q}$ become known variables, and delivered to Eq. (1) for closure. The coupling scheme of Eqs. (6) and (1) are detailed presented in Section 2.4.

2.2. Spatial discretization using VCFVM

We begin by partitioning the spatial domain Ω into an arbitrary number of non-overlapping polygons of various shapes, referred to as primary cells in our scheme, as illustrated in Fig. 1. The vertices and centers of the primary cells are denoted as $\boldsymbol{x}_v$ and $\boldsymbol{x}_K$, respectively, while the midpoints of the primary edges are designated as $\boldsymbol{x}_\sigma$. The dual cells are constructed by sequentially connecting the cell centers and edge midpoints. The dual cell also serves as the control volume for the vertex that contained within it. The primary unknowns for water flow and solute transport, i.e., H and c, are defined at the vertices. Additionally, for convenient of numerical flux construction, auxiliary unknowns are defined at the cell centers and edge midpoints, and they are computed via linear interpolation from the primary unknowns in the final scheme. It is important to note that for the spatial discretization of nonmatching vertices, the corresponding nonmatching interface is treated as a polyline where the nonmatching vertices serve as its vertices. For example, in Fig. 1(b), $\boldsymbol{x}_v$ is a nonmatching vertex, therefore, the primary cell K_1 is treated as a quadrilateral, rather than a triangle. Consequently, the spatial discretization scheme presented above can be directly applied to nonmatching grid without any technical difficulties.

2.2.1. Numerical diffusive flux in VCFVM

The diffusive flux of solute transport refers to the first term within the left-hand side parentheses, which is driven by the gradient of solute concentration and the hydrodynamic diffusion tensor. For illustration, we consider the cell K shown in Fig. 1(b) for instance. According to Fick's law, the diffusion flux $\boldsymbol{v}$ at any point within cell K is given by

$$\boldsymbol{v} = -\theta_K \boldsymbol{D}_K \nabla c, \tag{10}$$

where $\boldsymbol{D}_K$ denotes the constant approximation of the hydrodynamic dispersion tensor $\boldsymbol{D}$ within the primary cell K, taken as the value of $\boldsymbol{D}$ at the cell center; and θ_K represents the volumetric water content within cell K. Then, the diffusion outflow of the control

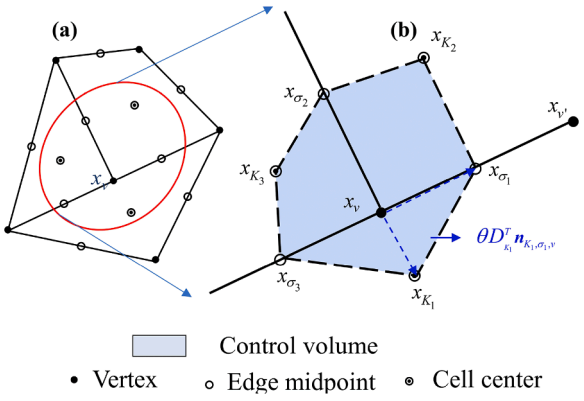


Fig. 1. Schematic diagram of VCFVM: (a) the primary cell (solid line), including vertices, edge midpoints, and cell centers; (b) the dual cell (dashed line) and notations for constructing the numerical diffusive flux. Note: $\boldsymbol{x}_\upsilon$, $\boldsymbol{x}_\sigma$, and $\boldsymbol{x}_K$ denote the specified vertex, the midpoint of the primary edge σ, and the center of the primary cell K, respectively; $\boldsymbol{D}_K$ represents the constant approximation of the hydrodynamic dispersion tensor $\boldsymbol{D}$ within the primary cell K, taken as the value of $\boldsymbol{D}$ at the cell center; $\boldsymbol{n}_{K,\sigma,\upsilon}$ refers to the unit normal vector (outward) to the edge $\boldsymbol{x}_\sigma\boldsymbol{x}_K$ of the control volume associated with the vertex $\boldsymbol{x}_\upsilon$.

volume centered at vertex υ crossing the edge $\boldsymbol{x}_\sigma\boldsymbol{x}_K$, denoted as $F^d_{K,\sigma,\upsilon}$, is the product of the length of the projection of the flux rate $\boldsymbol{v}$ along the normal $\boldsymbol{n}_{K,\sigma,\upsilon}$ to the edge $\boldsymbol{x}_\sigma\boldsymbol{x}_K$ and the length of the edge $|\boldsymbol{x}_\sigma\boldsymbol{x}_K|$, i.e,

$$F^d_{K,\sigma,\upsilon} = -|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K\nabla c\cdot\boldsymbol{n}_{K,\sigma,\upsilon} = -\nabla c\cdot\left(|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon}\right). \tag{11}$$

Furthermore, the vector $|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon}$ in Eq. (11) can be decomposed into a linear combination of two vectors $\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}$ and $\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}$, resulting in,

$$|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon} = \alpha_{K,\sigma,\upsilon}\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K} + \beta_{K,\sigma,\upsilon}\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}, \tag{12}$$

where

$$\alpha_{K,\sigma,\upsilon} = \frac{|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon}\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}^\perp}{\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}^\perp}, \tag{13a}$$

$$\beta_{K,\sigma,\upsilon} = \frac{|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon}\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}^\perp}{\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}^\perp}, \tag{13b}$$

and $\boldsymbol{v}^\perp$ denotes the vector obtained by rotating the vector $\boldsymbol{v}$ counterclockwise by 90 degrees. Supposing that the solute concentration c is linearly changed along $\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}$ and $\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}$, i.e., $\nabla c\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_K}\simeq(c_K-c_\upsilon)$, and $\nabla c\cdot\overrightarrow{\boldsymbol{x}_\upsilon\boldsymbol{x}_\sigma}\simeq(c_\sigma-c_\upsilon)$. Therefore, substituting Eq. (12) with Eq. (13) into Eq. (11) yields

$$F^d_{K,\sigma,\upsilon} \simeq \alpha_{K,\sigma,\upsilon}(c_\upsilon-c_K)+\beta_{K,\sigma,\upsilon}(c_\upsilon-c_\sigma), \tag{14}$$

where c_υ, c_K, and c_σ denote the solute concentrations at $\boldsymbol{x}_\upsilon$, $\boldsymbol{x}_K$, and $\boldsymbol{x}_\sigma$, respectively. The symbol $\simeq$ indicates that the relation preserves linearity-that is, it holds exactly when the unknown c is piecewise linear and the diffusion tensor $\boldsymbol{D}$ is piecewise constant within the primary cell K.

Similarly, the outflow of the control volume centered at vertex υ' crossing the edge $\boldsymbol{x}_\sigma\boldsymbol{x}_K$ can be approximated as follows:

$$F^d_{K,\sigma,\upsilon'} \simeq \alpha_{K,\sigma,\upsilon'}(c_{\upsilon'}-c_K)+\beta_{K,\sigma,\upsilon'}(c_{\upsilon'}-c_\sigma), \tag{15}$$

where

$$\alpha_{K,\sigma,\upsilon'} = \frac{|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon'}\cdot\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_\sigma}^\perp}{\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_K}\cdot\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_\sigma}^\perp}, \tag{16a}$$

$$\beta_{K,\sigma,\upsilon'} = \frac{|\boldsymbol{x}_\sigma\boldsymbol{x}_K|\theta_K\boldsymbol{D}_K^T\boldsymbol{n}_{K,\sigma,\upsilon'}\cdot\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_K}^\perp}{\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_\sigma}\cdot\overrightarrow{\boldsymbol{x}_{\upsilon'}\boldsymbol{x}_K}^\perp}. \tag{16b}$$

2.2.2. Numerical advection flux in VCFVM and treatments for numerical oscillation

The advection flux in solute transport corresponds to the second term inside the parentheses on the left-hand side of Eq. (1a), representing the solute carried by water flow. The advection outflow from the control volume centered at vertex υ across the edge $\boldsymbol{x}_\sigma\boldsymbol{x}_K$, denoted by $F^a_{K,\sigma,\upsilon}$, is given by

$$F^a_{K,\sigma,\upsilon} = \gamma_{K,\sigma,\upsilon}\tilde{c}_\sigma, \tag{17}$$

where $\tilde{c}_\sigma$ is the solute concentration related to the water flow $\gamma_{K,\sigma,\upsilon}$, and its definition varies among different numerical technics, which is detail introduced later in this subsection; $\gamma_{K,\sigma,\upsilon}$ is defined as

$$\gamma_{K,\sigma,\upsilon} = |\boldsymbol{x}_\sigma \boldsymbol{x}_K| \boldsymbol{q}_K \cdot \boldsymbol{n}_{K,\sigma,\upsilon}, \tag{18}$$

and $\boldsymbol{q}_K$ denotes the Darcy velocity vector within cell K. Note that in Eq. (17), the solute concentration associated with the water flow crossing $\boldsymbol{x}_\sigma \boldsymbol{x}_K$ is typically approximated by the solute concentration at the midpoint.

In numerical simulation, numerical oscillation usually occurs in advection-dominated region, and the relative intensity of advection term and diffusion term could be quantified by local Péclet number (Pe), which is defined as,

$$\mathrm{Pe} = \frac{|\boldsymbol{q}_K| \Delta x}{\theta D}, \tag{19}$$

where Δx is the spatial resolution. A larger value of Pe indicates a higher likelihood of numerical oscillations. Therefore, adopting refined grid could effectively mitigate numerical oscillation by reducing the spatial step size Δx in advection dominated domain, which in turn reduces local Pe.

Besides grid refinement, additional numerical strategies can be employed to suppress numerical oscillations in advection-dominated flow problems. Although these methods are not the focus of this study, comparative tests were conducted in Case 1 to evaluate the performance of the proposed method with grid refinement to three selected strategies. A brief introduction to these strategies is provided below.

First is upstream weighting method (**UW**, see [37]),which is a widely used technique for mitigating spurious oscillations in groundwater transport modeling. Rather than directly using the concentration at a point along the edge, this method computes the concentration based on the flow direction by selecting an appropriate value from the upstream vertex. This flow-aligned interpolation enhances numerical stability by incorporating the directionality of advection. Specifically, the interpolated concentration $\tilde{c}_\sigma$ in Eq. (17) is given by:

$$\tilde{c}_\sigma = (0.5 + 0.5\lambda\tau_{\upsilon,\upsilon'})c_\upsilon + (0.5 + 0.5\lambda\tau_{\upsilon',\upsilon})c_{\upsilon'}, \tag{20}$$

where the first and second parenthetical terms are the weights w_υ and $w_{\upsilon'}$ assigned to vertices υ and υ', respectively; λ is an empirical parameter (ranging from 0.0012 to 0.008, and is set as 0.005 in this study). The convection-dispersion ratios $\tau_{\upsilon,\upsilon'}$ and $\tau_{\upsilon',\upsilon}$ are defined as:

$$\tau_{\upsilon,\upsilon'} = \frac{q_{\upsilon,\upsilon'}|\boldsymbol{x}_\upsilon \boldsymbol{x}_{\upsilon'}|}{\alpha_L |\boldsymbol{q}_K|}, \quad \tau_{\upsilon',\upsilon} = \frac{q_{\upsilon',\upsilon}|\boldsymbol{x}_\upsilon \boldsymbol{x}_{\upsilon'}|}{\alpha_L |\boldsymbol{q}_K|}, \tag{21}$$

where $q_{\upsilon,\upsilon'}$ is the projection of Darcy velocity vector on direction $\boldsymbol{x}_\upsilon \boldsymbol{x}_{\upsilon'}$.

The second is time step limiting method (**TL**), which suppresses numerical oscillations indirectly by constraining the time step size Δt. The TL strategy was designed for Galerkin finite element method [5,38,39], in which a constraint on the product of the Pe and Courant number (Cr) is introduced to mitigate numerical oscillations, i.e.,

$$\mathrm{Pe} \cdot \mathrm{Cr} \le \omega, \tag{22}$$

where

$$\mathrm{Cr} = \frac{|\boldsymbol{q}_K| \Delta t}{\theta \Delta x}. \tag{23}$$

Here, Δx is the spatial resolution, and ω is a user-defined threshold (typically less than 2.0; set to 1.0 in this study). Substituting the above expressions leads to a condition on the time step:

$$\Delta t \le \frac{\omega \theta^2 D}{|\boldsymbol{q}_K|^2}. \tag{24}$$

Though the **TL** has been well demonstrated to be effective for Galerkin finite element method, its effectiveness in our proposed scheme still remain to be evaluated. Therefore, its effectiveness in mitigating numerical oscillations is evaluated in Case 1.

The last selected strategy is MUSCL-type approximation approach with a Multidimensional Limiting Process ([14,40]), denoted as **MLP** here. In the **MLP** option, the solute concentration $\tilde{c}_\sigma$ associated with the water flow $\gamma_{K,\sigma,\upsilon}$ crossing a given dual edge $\boldsymbol{x}_\sigma \boldsymbol{x}_K$ is approximated using the following formula:

$$\tilde{c}_\sigma = L_{fd} c_{\upsilon,e} + (1 - L_{fd}) c_{\upsilon',e}, \tag{25}$$

where L_{fd} is a logical variable, which equal to 1 if $\gamma_{K,\sigma,\upsilon} > 0$, and equal to 0 for other cases; $c_{\upsilon,e}$ and $c_{\upsilon',e}$ are defined as,

$$c_{\upsilon,e} = c_\upsilon - \frac{\Psi^*_\upsilon}{4}((1-k)\Delta c_\upsilon + (1+k)(c_\upsilon - c_{\upsilon'})), \tag{26a}$$

$$c_{\upsilon',e} = c_{\upsilon'} - \frac{\Psi^*_{\upsilon'}}{4}((1-k)\Delta c_{\upsilon'} + (1+k)(c_\upsilon - c_{\upsilon'})), \tag{26b}$$

where k is a user defined parameter to control the degree of approximation, and set as $\frac{1}{3}$ in this study [14]; Ψ^*_υ and $\Psi^*_{\upsilon'}$ are the so-called limiter function at vertices $\boldsymbol{x}_\upsilon$ and $\boldsymbol{x}_{\upsilon'}$, respectively; the upstream and downstream operators Δc_υ and $\Delta c_{\upsilon'}$ are defined by

$$\Delta c_\upsilon = 2\nabla c_\upsilon \cdot \overrightarrow{\boldsymbol{x}_\upsilon \boldsymbol{x}_{\upsilon'}} - (c_{\upsilon'} - c_\upsilon), \tag{27a}$$

$$\Delta c_{v'} = 2\nabla c_{v'} \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_{v'}} - (c_{v'} - c_v), \tag{27b}$$

where the solute gradient ∇c_v is approximated using Least-Square approach among the concentration of the vertices neighbor to the vertex $\boldsymbol{x}_x$, and similar for $\nabla c_{v'}$.

In MLP, the limiter function Ψ_v^* is obtained by:

$$\Psi_v^* = \min_{j \in \mathcal{V}_{in}} \left(\Psi_{j,v}\right), \tag{28a}$$

$$\Psi_{j,v} = \begin{cases} \min(1, r_{j,v}), \nabla c_v \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_j} > 0 \\ 1, \text{otherwise} \end{cases} \tag{28b}$$

where $\mathcal{V}_{in}$ denotes the set of interpolation points (i.e., $\boldsymbol{x}_\sigma$ and $\boldsymbol{x}_K$) neighbor to vertex $\boldsymbol{x}_v$. The limiter function function $\Psi_{v'}^*$ is calculated suing similar method.

In Eq. (28b), $r_{j,v}$ is calculated using:

$$r_{j,v} = \begin{cases} \dfrac{c_{j,\max} - c_v}{\nabla c \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_j}}, \nabla c_v \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_j} > 0 \\ \dfrac{c_{j,\min} - c_v}{\nabla c \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_j}}, \text{otherwise} \end{cases} \tag{29}$$

where $c_{j,max}$ and $c_{j,min}$ are the minimum and maximum concentrations among the points in $\mathcal{V}_{in}$.

Combining Eq. (25) to Eq. (29) leads to:

$$\tilde{c}_\sigma = A_{MLP} c_v + B_{MLP} c_{v'} + C_{MLP}, \tag{30}$$

where A_{MLP}, B_{MLP} and C_{MLP} are scalar coefficient defined as:

$$A_{MLP} = L_{fd} - \frac{L_{fd}\Psi_v^*}{6} + \frac{\Psi_{v'}^*}{6} - \frac{L_{fd}\Psi_{v'}^*}{6}, \tag{31a}$$

$$B_{MLP} = 1 - L_{fd} - \frac{\Psi_v^*}{6} + \frac{L_{fd}\Psi_{v'}^*}{6} + \frac{L_{fd}\Psi_v^*}{6}, \tag{31b}$$

$$C_{MLP} = \frac{L_{fd}\Psi_v^*}{3} \nabla c_v \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_{v'}} - (1 - L_{fd}) \frac{\Psi_{v'}^*}{3} \nabla c_{v'} \cdot \overrightarrow{\boldsymbol{x}_v \boldsymbol{x}_{v'}}. \tag{31c}$$

Among the three technics mentioned technics mentioned above, the **TL** method, does not modify the interpolation scheme for $\tilde{c}_\sigma$ the **UW** and **MLP** method. The **TL** can be used either in conjunction with previously introduced interpolation schemes-such as those defined in Eq. (20) or Eq. (30)-or one may simply adopt a linear interpolation:

$$\tilde{c}_\sigma = \frac{1}{2}(c_v + c_{v'}). \tag{32}$$

2.2.3. Numerical flux in VCFVM

Combining the diffusion and advection terms, the total numerical flux across edge $\boldsymbol{x}_\sigma \boldsymbol{x}_K$ from the control volume centered at vertex v is given by:

$$F_{K,\sigma,v} = F_{K,\sigma,v}^d + F_{K,\sigma,v}^a = \alpha_{K,\sigma,v}(c_v - c_K) + \beta_{K,\sigma,v}(c_v - c_\sigma) + \gamma_{K,\sigma,v}\tilde{c}_\sigma. \tag{33}$$

where $\alpha_{K,\sigma,v}, \beta_{K,\sigma,v}$ and $\gamma_{K,\sigma,v}$ are defined in Eqs. (13a), (13b) and (18), respectively.

To eliminate auxiliary unknowns and construct the numerical scheme using only primary unknowns, the c_K and c_σ in diffusive flux are computed via linear interpolation from the primary unknowns as follows:

$$c_K = \frac{1}{n_K} \sum_{v \in \mathcal{V}_K} c_v, \tag{34a}$$

$$c_\sigma = \frac{1}{2}(c_v + c_{v'}), \tag{34b}$$

where $\mathcal{V}_K$ and n_K is the set and number of vertices in primary cell K. Substituting Eq. (34) into Eq. (33) lead to the final numerical fluxes, still denoted as $F_{K,\sigma,v}$ and $F_{K,\sigma,v'}$, and they are strictly local conservative [35], i.e.,

$$F_{K,\sigma,v} + F_{K,\sigma,v'} = 0, \tag{35}$$

which is an important characteristic that must be satisfied when constructing finite volume schemes.

2.2.4. Final spatial discretization scheme based on VCFVM

The total outflow F_v around the control volume K_v^* is the sum of the outflows across all of its dual edges. For an interior vertex, the left-hand side of the governing equation Eq. (1a) can be discretized as:

$$\nabla \cdot (\theta \boldsymbol{D} \nabla c - \boldsymbol{q} c) \approx \sum_{K \in \mathcal{M}_v} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_K} F_{K,\sigma,v}, \tag{36}$$

where $\mathcal{M}_v$ and $\mathcal{E}_v$ denote the set of primary cells and edges sharing vertex $\boldsymbol{x}_v$, respectively, and $\mathcal{E}_K$ is the set of primary edges of cell K. The area of the control volume K_v^* is denoted as $|K_v^*|$. It should be noted that the in Eq. (36), the numerical fluxes $F_{K,\sigma,v}$ across different dual edges are calculated separately without lumping.

On the right-hand side of the governing equation Eq. (1a), the discretization for the integral over the control volume K_v^* is:

$$\int_{K_v^*} \left(\frac{\partial \theta c}{\partial t} + f(\boldsymbol{x}, t) \right) dx \approx |K_v^*| \left(\frac{\partial \theta_v c_v}{\partial t} + f(\boldsymbol{x}_v, t) \right). \tag{37}$$

To summarize, the spatial discretization of the governing Eq. (1a) for a given inner vertex in matrix form is:

$$\boldsymbol{B}\boldsymbol{c} = \boldsymbol{\Lambda} \left(\frac{\partial (\boldsymbol{\theta} \circ \boldsymbol{c})}{\partial t} + \boldsymbol{f}(t) \right), \tag{38}$$

where $\boldsymbol{B}$ is the stiffness matrix, which can be expressed as:

$$\boldsymbol{B} = \boldsymbol{B}^d + \boldsymbol{B}^a, \tag{39}$$

where $\boldsymbol{B}^d$ and $\boldsymbol{B}^a$ are matrices related to diffusion and advection respectively, and their entries are defined as:

$$B^d_{v,v'} = \begin{cases} \displaystyle\sum_{K \in \mathcal{M}_v} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_K} \left(\alpha_{K,\sigma,v} - \frac{\alpha_{K,\sigma,v}}{N_K} + \frac{\beta_{K,\sigma,v}}{2} \right), & v = v', \\ \displaystyle -\sum_{K \in \mathcal{M}_v \cap \mathcal{M}_{v'}} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_{v'}} \frac{\beta_{K,\sigma,v}}{2} - \sum_{K \in \mathcal{M}_v \cap \mathcal{M}_{v'}} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_K} \frac{\alpha_{K,\sigma,v}}{N_K}, & v \neq v' \text{ and } v' \notin \mathcal{V}_D, \end{cases}$$

$$B^a_{v,v'} = \begin{cases} \displaystyle\sum_{K \in \mathcal{M}_v} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_K} (\omega_v \gamma_{K,\sigma,v}), & v = v', \\ \displaystyle\sum_{K \in \mathcal{M}_v \cap \mathcal{M}_{v'}} \sum_{\sigma \in \mathcal{E}_v \cap \mathcal{E}_{v'}} (1 - w_v) \gamma_{K,\sigma,v}, & v \neq v' \text{ and } v' \notin \mathcal{V}_D, \end{cases}$$

where $\boldsymbol{\theta} = \{\theta_v,\ v \in \mathcal{V} \backslash \mathcal{V}_D\}$, $\boldsymbol{c} = \{c_v,\ v \in \mathcal{V} \backslash \mathcal{V}_D\}$ and $\boldsymbol{\theta} \circ \boldsymbol{c}$ denotes the element-wise product between $\boldsymbol{\theta}$ and $\boldsymbol{c}$; $\boldsymbol{\Lambda}$ is a diagonal matrix whose diagonal entries are $\{|K_v^*|, v \in \mathcal{V} \backslash \mathcal{V}_D\}$. The default value of the upwind weight w_v is set as 0.5 in this study, and is calculated based on Eq. (20) if upstream weight treatment is activated.

2.3. Treatments of boundary conditions and sink/source terms

There are two kinds of boundary conditions incorporated in the proposed algorithm, namely Dirichlet boundary and Neumann boundary, as defined in Eqs. (1c) and (1d). For the vertices on Dirichlet boundary (hereinafter referred to as Dirichlet vertices), since the unknown variable (i.e., solute concentration) at them is prescribed, the solute transport equation corresponding to them are excluded from the equation system. In other words, the number of the row of the final equation (n_{row}) system is:

$$n_{row} = n_v - n_{vD}, \tag{40}$$

where n_v and n_{vD} are the number of the vertices in the spatial domain and on the Dirichlet boundaries, respectively. For vertices directly influenced by Dirichlet vertices, i.e., those share common primary cells with Dirichlet vertices, their terms in the final spatial discretization scheme that related to the Dirichlet vertices are moved to the right-hand side vector of Eq. (38).

For the vertices on the Neumann boundary (hereinafter referred to as Neumann vertices), their concentrations are unknown, therefore, they are included in the equation system. Besides, the Neumann boundary flux is assumed to be normal to the boundary, with the value assumed to be constant within the governing length of each Neumann vertex (i.e., $l_{v,N}$). Then the numerical flux across the Neumann boundary could be expressed as $g_N l_{v,N}$, and is added in the corresponding right-hand side vector of Eq. (38).

In practical applications, sink/source terms in the solute transport field, which are common in agricultural systems, need to be specifically considered. In our algorithm, the general sink/source term f_v for a given vertex v is defined as:

$$f_v = w_v c_w, \tag{41}$$

where w_v is the amount of sink/source water applied to vertex v, with units of $[L^2T^{-1}]$; c_w is the solute concentration of the sink/source water, with units of $[ML^{-3}]$.

For test case 2, the values of w_v and c_w for the drippers are preset before the simulation begins, while for test case 3, the values of w_v and c_w for the subsurface drainage pipe at the $n+1$ time step are calculated using the following formulas [15,41]:

$$w_v^{n+1} = \begin{cases} 0, & h_{sdr}^{n+1} \le D_e, \\ C_d(h_{sdr}^{n+1} - D_e)l_{sdr}, & h_{sdr}^{n+1} > D_e, \end{cases} \tag{42}$$

$$c_w^{n+1} = c_{sdr}^n, \tag{43}$$

where h_{sdr} is the groundwater level at the subsurface pipe vertex, [L]; D_e is the burial depth of the subsurface pipe, [L]; l_{sdr} is the length of the subsurface pipe within the control volume, [L]; C_d is the resistance coefficient, with unit of [LT^{-1}], and c_{sdr} is the solute concentration at the vertex where the subsurface pipe is located, [ML^{-3}].

To incorporate the treatments for Dirichlet and Neumann boundary conditions, the matrix form defined in (38) should be modified as:

$$\boldsymbol{B}\boldsymbol{c} = \boldsymbol{\Lambda}\left(\frac{\partial(\boldsymbol{\theta}\circ\boldsymbol{c})}{\partial t} + \boldsymbol{f}(t)\right) + \tilde{\boldsymbol{b}}, \tag{44}$$

where $\tilde{\boldsymbol{b}}$ is the vector reflect the influence of the boundary vertices, whose entries are

$$\tilde{b}_v = \tilde{b}_{v,D} + \tilde{b}_{v,N}, \tag{45}$$

where $\tilde{b}_{v,D}$ and $\tilde{b}_{v,N}$ are entries corresponding to the Dirichlet vertex and Neumann vertex. They are defined as,

$$\begin{aligned} \tilde{b}_{v,D} =& g_D \sum_{v'\in\mathcal{V}_D}\left(-\sum_{K\in\mathcal{M}_v\cap\mathcal{M}_{v'}}\sum_{\sigma\in\mathcal{E}_v\cap\mathcal{E}_{v'}}\frac{\beta_{K,\sigma,v}}{2} - \sum_{K\in\mathcal{M}_v\cap\mathcal{M}_{v'}}\sum_{\sigma\in\mathcal{E}_v\cap\mathcal{E}_K}\frac{\alpha_{K,\sigma,v}}{N_K}\right) \\ &+ g_D\sum_{v'\in\mathcal{V}_D}\left(\sum_{K\in\mathcal{M}_v\cap\mathcal{M}_{v'}}\sum_{\sigma\in\mathcal{E}_v\cap\mathcal{E}_K}(1-w_v)\gamma_{K,\sigma,v}\right), \end{aligned} \tag{46a}$$

$$\tilde{b}_{v,N} = -g_N l_{v,N}, v \in \mathcal{V}_N. \tag{46b}$$

2.4. Temporal discretization scheme

To obtain a fully discretized numerical scheme, we also need to perform temporal discretization on the semi-discrete system. To do this, we first partition the temporal domain $[0,T]$ into N intervals, with the length of the nth time interval denoted as τ^n. Next, we employ the Backward Euler method and Crank-Nicolson for temporal discretization. By using Backward Euler method for temporal discretization, Eq. (38) is discretized as:

$$\boldsymbol{B}^{n+1}\boldsymbol{c}^{n+1} = \boldsymbol{\Lambda}\left(\frac{\boldsymbol{\theta}^{n+1}\circ\boldsymbol{c}^{n+1} - \boldsymbol{\theta}^n\circ\boldsymbol{c}^n}{\tau^{n+1}} + \boldsymbol{f}^{n+1}\right), \tag{47}$$

where the superscript $n+1$ and n represent the current and previous time levels; $\boldsymbol{f}$ is a source/sink term vector of which the mth element is f_m.

For Crank-Nicolson method, Eq. (38) is discretized as:

$$\frac{1}{2}\left(\boldsymbol{B}^{n+1}\boldsymbol{c}^{n+1} + \boldsymbol{B}^n\boldsymbol{c}^n\right) = \boldsymbol{\Lambda}\left(\frac{\boldsymbol{\theta}^{n+1}\circ\boldsymbol{c}^{n+1} - \boldsymbol{\theta}^n\circ\boldsymbol{c}^n}{\tau^{n+1}} + \boldsymbol{f}^{n+1}\right). \tag{48}$$

Eqs. (47) and (48) can be organized into a linear system of equations, which is written as:

$$\boldsymbol{A}\boldsymbol{c}^{n+1} = \boldsymbol{b}, \tag{49}$$

where for Backward Euler method,

$$\boldsymbol{A} = \frac{1}{\tau^{n+1}}\mathrm{diag}(\boldsymbol{\Lambda}\boldsymbol{\theta}^{n+1}) - \boldsymbol{B}^{n+1}, \tag{50}$$

$$\boldsymbol{b} = \frac{1}{\tau^{n+1}}\boldsymbol{\Lambda}(\boldsymbol{\theta}^n\circ\boldsymbol{c}^n) - \boldsymbol{\Lambda}\boldsymbol{f}^{n+1}. \tag{51}$$

And for Crank-Nicolson method,

$$\boldsymbol{A} = \frac{2}{\tau^{n+1}}\mathrm{diag}(\boldsymbol{\Lambda}\boldsymbol{\theta}^{n+1}) - \boldsymbol{B}^{n+1}, \tag{52}$$

$$\boldsymbol{b} = \frac{2}{\tau^{n+1}}\boldsymbol{\Lambda}(\boldsymbol{\theta}^n\circ\boldsymbol{c}^n) - 2\boldsymbol{\Lambda}\boldsymbol{f}^{n+1} + \boldsymbol{B}^n\boldsymbol{c}^n. \tag{53}$$

Eq. (49) represents the fully discretized spatial-temporal scheme. The final coefficient matrix $\boldsymbol{A}$ is asymmetric, spares in any case, and diagonally dominated in diffusion dominated scenarios. At each time step, $\boldsymbol{A}$ is assembled based on Eqs. (50) or (52). To enhance

compute efficient, for groundwater solute transport, where θ and $\boldsymbol{B}^d$ are time-independent, $\boldsymbol{B}^d$ is calculated only one time before simulation starting, and used directly to assemble $\boldsymbol{A}$ at every time step. The Generalized Minimum Residual (GMRES) method with Incomplete LU Decomposition Preconditioner (ILU) is adopted to solve the equation system defined in Eq. (49). Then the solute concentration at any given point in the spatial-temporal domain can be obtained by solving this linear system, without the need for additional iteration.

2.5. Coupling scheme between water and solute module

In the proposed solute transport algorithm (solute module), the water flow, which is the driver of solute transport, is solved in the water module based on the VCFVM. These two modules are coupled by delivering $\boldsymbol{q}$ and θ from the water module to the solute module within each time step. The detailed coupling scheme between water and solute modules are presented in Algorithm 1 for clarity. It should be noted that both Eqs. (5) and (6) are nonlinear equations, thus linearization and iteration are required within each time step, and the technical details are clarified in [34,35].

Algorithm 1: Water and solute modules coupling scheme.

Input: The aquifer hydraulic parameters involved in Eq. (5) or soil hydraulic parameters involved in Eq. (6), the initial value g_0, the boundary values g_1 and g_2, and the spatial mesh $\mathcal{M}$

```
Compute the geometric and topological information of each vertex;
Set the initial conditions, boundary conditions and parameters to the prescribed locations;
for time t = t_1, t_2, ··· do
    if The concerned scenario is in unconfined aquifer then
        for iteration step i = 1, 2, ··· do
            Solving linearized equation system of Eq. (5);
            if the convergence criterion is satisfied then
                Exit
    else
        for iteration step i = 1, 2, ··· do
            Solving linearized equation system of Eq. (6);
            if the convergence criterion is satisfied then
                Exit
    Compute the Darcy flow q of each cell based on Eq. (3) and delivered to solute module with θ;
    Calculate spatial discretized coefficient matrix B;
    Assemble coefficient matrix A and vector b after temporal discretization;
    Solve the equation system Eq. (49);
```

Output: Water potential h and solute concentration c at given spatial-temporal points.

3. Numerical experiments setting

In this study, we implement the proposed scheme in C++ and execute it on Ubuntu 18.04 with GCC 7.3.0. To assess the accuracy and efficiency of the scheme in simulating solute transport with high local solute gradients, we evaluate it using three types of test cases: a synthetic case, a small-scale case, and a field-scale case. Efficiency is measured based on CPU time across different meshes, while accuracy is quantified by comparing the simulated results to benchmark values. We use two error indices: discrete L_1-norm error and relative L_2-norm error, denoted as E_1 and RE_2 [42]. The are defined as follows:

$$E_1 = \frac{\sum_{v\in\mathcal{V}} |K_v^*||c_v - c_v^*|}{\sum_{v\in\mathcal{V}} |K_v^*|}, \tag{54}$$

$$RE_2 = \sqrt{\frac{\sum_{v\in\mathcal{V}} |K_v^*|\left(\frac{c_v - c_v^*}{c_v^*}\right)^2}{\sum_{v\in\mathcal{V}} |K_v^*|}} \times 100\%, \tag{55}$$

where c^* denotes the benchmark value, $\mathcal{V}$ represents the set of vertices used in the calculation; and $|K_v^*|$ denotes the area of the control volume of the corresponding vertex.

3.1. Case 1: Synthetic case

The first test case is a synthetic benchmark originally proposed in [43], used here to: (1) evaluate the spatial and temporal accuracy order of the VCFVM; (2) assess the accuracy of the VCFVM on nonmatching grids under varying Peclet numbers, and (3) evaluate

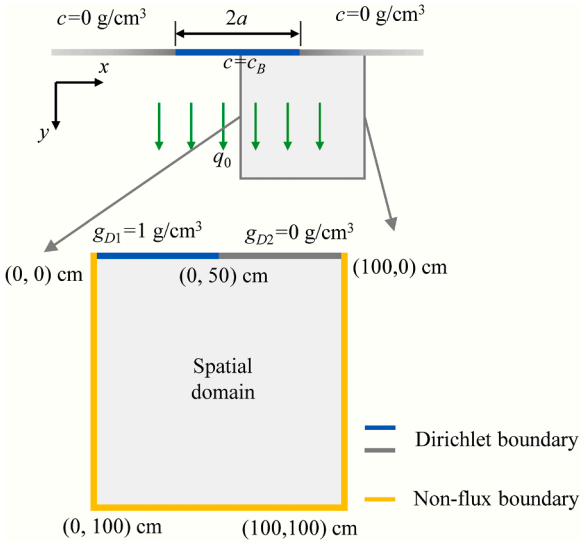


Fig. 2. Schematic diagram of the settings of Case 1. Note: g_{D1} and g_{D2} represent the values of the Dirichlet boundary conditions; q_0 is the Darcy velocity of water; c_B is the solute concentration on the surface.

the effectiveness of local grid refinement in mitigating numerical oscillations compared to conventional methods. The computational domain is assumed to be infinite in both the lateral and vertical (downward) directions, with a steady Darcy flow driven by a constant Darcy velocity q_0. Initially, the solute concentration throughout the domain is uniformly zero. A recharge zone of width $2a$ is located on the upper boundary, where the solute concentration is fixed at a constant value c_B; the remaining part of the upper boundary is maintained at zero concentration. The configuration of Case 1 is illustrated in Fig. 2. In this setup, the solute concentration at any spatiotemporal point (x, y, t) is given by the following analytical solution:

$$c(x,y,t) = \frac{c_0 y}{4\sqrt{\pi D_{yy}}} \exp\left(\frac{q_0 y}{2D_{yy}}\right) \cdot \int_0^t \exp\left(-\frac{q_0^2\tau}{4D_{yy}} - \frac{y^2}{4D_{yy}\tau}\right)\tau^{-3/2}\left[\operatorname{erf}\left(\frac{a-x}{2\sqrt{D_{xx}\tau}}\right) + \operatorname{erf}\left(\frac{a+x}{2\sqrt{D_{xx}\tau}}\right)\right] d\tau. \tag{56}$$

Note that in Eq. (56), the coordinate origin is set at the midpoint of the recharge area on the upper boundary.

For the numerical simulation, the domain $\Omega = [0, 100]\,\text{cm} \times [0, 100]\,\text{cm}$ is chosen, as depicted by the gray shaded region in Fig. 2, with the temporal domain defined as $T = [0, 365]\,\text{d}$. The origin of the coordinate system coincides with the upper-left corner of the domain. Dirichlet boundary conditions are applied along the top boundary, while the left boundary is set as a no-flux boundary due to symmetry in solute transport across the domain. The bottom and right boundaries are also no-flux, as the numerical flux near these boundaries is negligible given the chosen parameters and spatiotemporal domain. The initial and boundary conditions are summarized as follows:

$$\begin{aligned} &c(\boldsymbol{x}, 0) = 0\ \text{g/cm}^3,\ \ \boldsymbol{x} \in \Omega, \\ &c(\boldsymbol{x}, t) = \begin{cases} c_B, & 0 \le x \le a,\ y = 0\ \text{cm}, \\ 0\ \text{g/cm}^3, & x > a,\ y = 0\ \text{cm}, \end{cases} \\ &(\theta \boldsymbol{D}\nabla c - \boldsymbol{q}c)\cdot \boldsymbol{n} = 0,\ \ x = 0\ \text{cm or } x = 100\ \text{cm or } y = 100\ \text{cm}. \end{aligned} \tag{57}$$

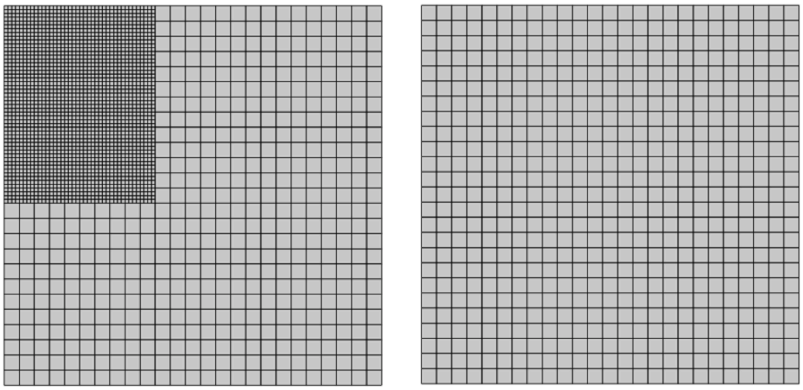
To evaluate the accuracy order, two different values of D_{xx} and D_{yy} are set, i.e., 0.01 cm^2/d and 0.1 cm^2/d, to evaluate the performance of the proposed algorithm in advection dominant and diffusion dominant scenarios, respectively. The boundary conditions involved in Eq. (57) are listed in Table 1. Here both normal VCFVM (i.e., VCFVM without additional treatment on advection flux and adopting Backward Euler for temporal discretization), and VCFVM with **MLP**, are selected to evaluate the spatial accuracy order, and both normal VCFVM and VCFVM with Crank-Nicolson are selected for temporal accurate order evaluation. To assess spatial accuracy order, four different spatial step sizes, i.e., 1.25, 0.625, 0.3125 and 0.15625 cm, are employed with time step size being a constant with a value of 2 d. To evaluate temporal accuracy order, four different time step sizes, i.e., 4, 2, 1 and 0.5 d, are adopted with spatial step size being a constant with a value of 1.25 cm.

To comprehensively evaluate the accuracy of the VCFVM under various Peclet numbers, more values of D_{xx} and D_{yy} are set to construct scenarios ranging from diffusion-dominated to advection-dominated, labeled as S_1 to S_4. As discussed in Section 2.1.2, numerical oscillations are a common phenomenon under advection-dominated conditions. To mitigate such oscillations, we apply numerical treatments such as local grid refinement (**LGR**), **UW**, **TL**, and **MLP**. Note that here the nonmatching grids are employed for **LGR**. In this section, we select the advection-dominated scenario S_4 as a representative case for evaluation. As a benchmark,

Table 1
Parameter settings for the solute transport equation in Case 1.

Parameters		Values
a		50 cm
q_0		0.1 cm/d
c_B		1 g/cm^3
$D_{xx}(D_{yy})$	S_1	1 cm^2/d
	S_2	0.1 cm^2/d
	S_3	0.05 cm^2/d
	S_4	0.025 cm^2/d

Note: D_{xx} and D_{yy} represent the hydrodynamic dispersion coefficients in xx and yy directions, respectively.



(a) $\mathcal{M}_1^a$ (Local Refined grid) (b) $\mathcal{M}_1^b$ (Coarse grid)

Fig. 3. Two types of grids used in Case 1.

we perform simulations on a coarse grid without any numerical treatments using nromal VCFVM, denoted as Check (**CK**). Then, we employ the **LGR**, **UW**, **TL**, and **MLP** methods in combination with VCFVM for simulations. All results are compared with the benchmark solution to verify the advantages of LGR in mitigating numerical oscillations. The coarse grid used is labeled as $\mathcal{M}_2^a$, and the grids used in this test case are shown in Fig. 3, with their basic information summarized in Table 2.

3.2. Case 2: Field drip irrigation scenario

This case represents a small-scale real-world application, with a more complex setup than Case 1, incorporating transient flow and source terms. It is designed to evaluate the accuracy of VCFVM in capturing local high solute gradients induced by source terms and to assess its performance on various nonmatching grids. The case is based on a field drip irrigation experiment conducted in the Hetao Irrigation District of Inner Mongolia, China (108.29°E, 41.17°N). During the experiment, soil moisture and salinity were monitored. The soil moisture data were used as benchmarks in [35], while the soil salinity measurements are adopted in this study for model validation.

For completeness, a brief overview of the experimental setup is provided here, and more details about the site and treatments can be found in [35]. The plant pattern with mulched drip irrigation is shown in Fig. 4(a). Drip tapes were spaced 1.0 m apart and placed at the center of plastic films with a width of 0.6 m. The drippers on each tape were spaced 30 cm apart. Maize was planted 20 cm away from the drip tape on both sides. During the experiment, the groundwater table was approximately 2.1 m deep, and the average discharge rate of each dripper was 777 cm^3/h. The introduced solute concentration was negligible.

The selected soil profile, indicated by the shaded area in Fig. 4(b), was sampled at three time points: before, during, and after irrigation, denoted as $T_0 = 0$ d, $T_1 = 0.13$ d, and $T_2 = 0.53$ d, respectively. The sampling scheme for each time point is also illustrated in Fig. 4(b). Three vertical profiles were taken at different lateral distances from the dripper: directly beneath the dripper (P_1), 25 cm away (P_2), and 50 cm away (P_3). The observed values at T_0 are used as initial conditions, while those at T_1 and T_2 serve as benchmarks for model evaluation.

In this case, soil water flow and solute transport are simulated simultaneously. The simulation domain is defined based on the sampling profile, with the spatial domain set as $\Omega = [0, 100]$ cm $\times$ $[0, 210]$ cm and the temporal domain as $T = [0, 0.53]$ d, corresponding to the experimental period from T_0 to T_2. The soil water flow module and its boundary conditions are consistent with those described

Table 2
Details of the meshes used in each test case and their corresponding CPU times.

Case	Mesh	Number of vertices	Number of cells	CPU time (s)
Case 1	$\mathcal{M}_1^a$	2695	2575	48.6
	$\mathcal{M}_1^b$	676	625	8.9(**CK**) 9.7(**UW**) 17.3(**TL**) 11.1 (**MLP**)
Case 2	$\mathcal{M}_2^a$	1106	2095	20.0
	$\mathcal{M}_2^b$	61	44	0.8
	$\mathcal{M}_2^c$	61	60	0.8
	$\mathcal{M}_2^d$	138	110	2.1
	$\mathcal{M}_2^e$	391	344	6.7
Case 3	$\mathcal{M}_3^a$	7521	7344	45.5
	$\mathcal{M}_3^b$	3285	2754	21.5

Note: **CK**, **UW**, **TL** and **MLP** represent check, upstream weighting method, time step limiting method, and MUSCL-type approximation approach with a Multidimensional Limiting Process, respectively.

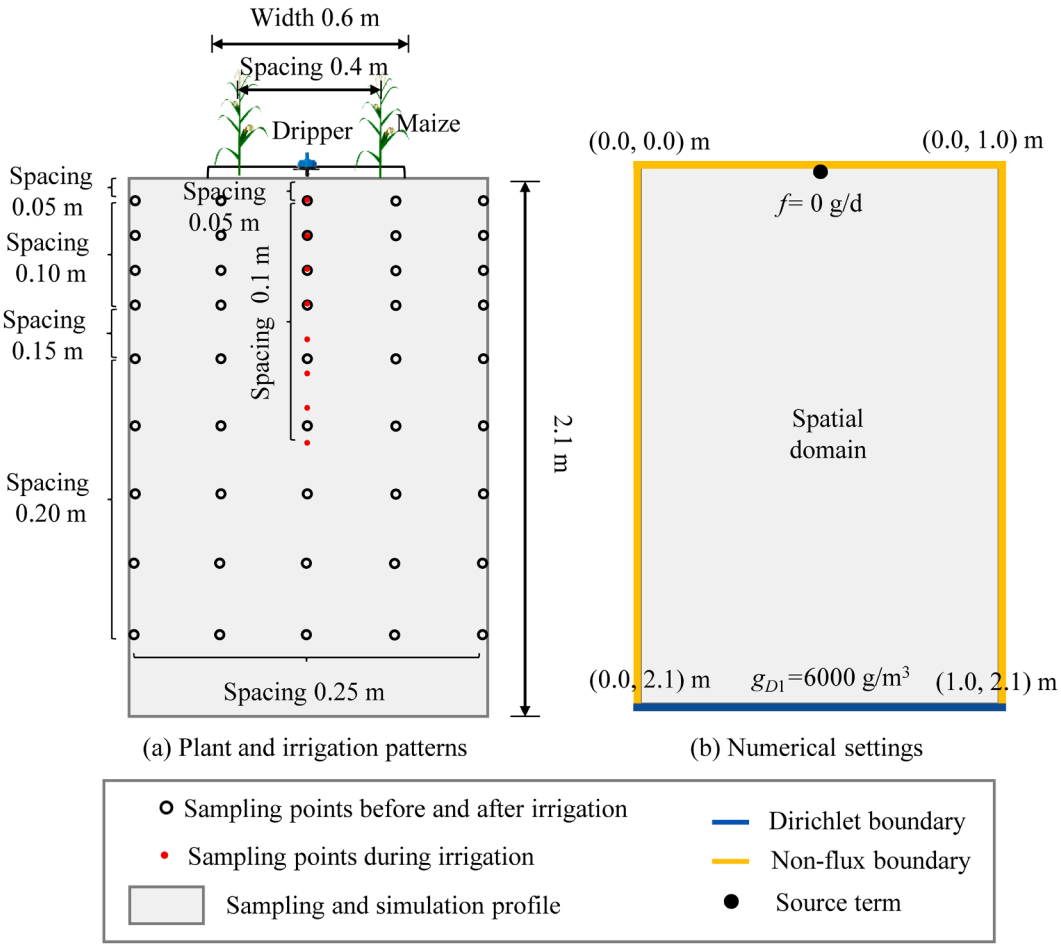


Fig. 4. Schematic diagram of Case 2: (a) plant and irrigation patterns of the experimental site, (b) numerical settings. Note: g_{D1} is the values of the Dirichlet boundary condition, and f represents the source term.

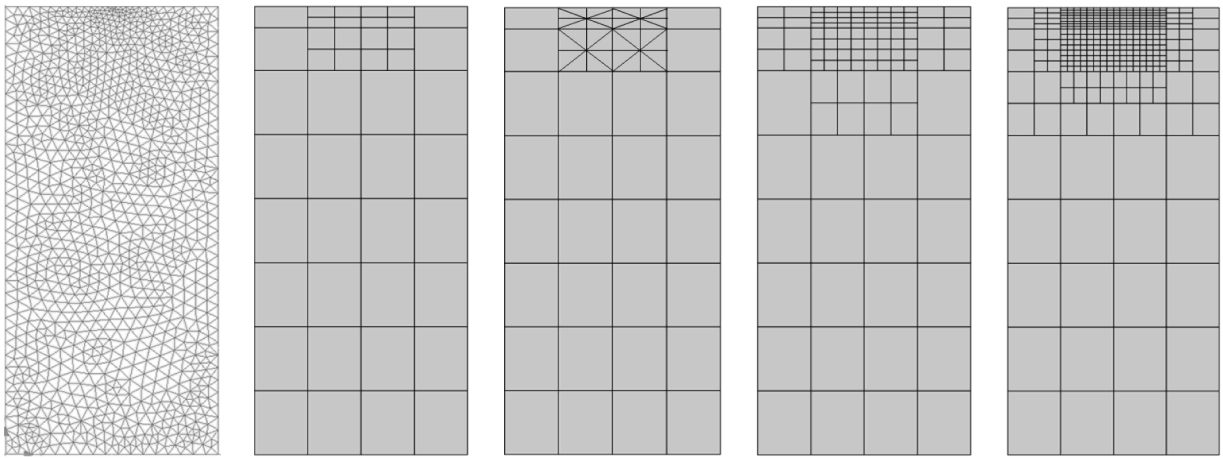


Fig. 5. Meshes used in Case 2, including the fine mesh $\mathcal{M}_2^a$ and several nonmatching grids $\mathcal{M}_2^b - \mathcal{M}_2^e$.

in Qian et al. [35], while the boundary conditions for solute transport are illustrated in Fig. 4 and are detailed as follows:

$$\begin{aligned} &c(\boldsymbol{x},t) = 0.006 \text{ g/cm}^3, \ y = 210 \text{ cm}, \\ &(\theta \boldsymbol{D}\nabla c - \boldsymbol{q}c)\cdot \boldsymbol{n} = \begin{cases} 0 \text{ g/cm}, & y = 0 \text{ cm}, \\ 0 \text{ g/cm}, & x = 0 \text{ or } 100 \text{ cm}, \end{cases} \\ &f(\boldsymbol{x},t) = 0 \text{ g/d}, \ \boldsymbol{x} = (50,0) \text{ cm}. \end{aligned} \tag{58}$$

Note that the bottom boundary is set as a Dirichlet boundary condition because it represents the groundwater level, where the concentration is expected to remain constant throughout the experiment. Additionally, although the solute introduced by the dripper can be considered negligible, it is still treated as a source term (see Fig. 4) due to the significant concentration gradient created by the water flow near the dripper. The values of the soil hydraulic parameters are the same as those in [35], while the solute transport parameters, including soil bulk density, α_L and α_T, are set to 1.45 g/cm^3, 20 cm, and 10 cm, respectively.

The case was initially simulated using the VCFVM on a nonmatching grid with a high refinement level, as illustrated in Fig. 5(d). The results were compared against observed values and the simulated results from HYDRUS on a fined triangular mesh, denoted as $\mathcal{M}_2^a$, to demonstrate the accuracy of VCFVM in a drip irrigation scenario. Subsequently, nonmatching grids with varying refinement levels and shapes were employed to test the applicability of VCFVM in handling arbitrary meshes, as shown in Fig. 5(b)–(e). Notably, among these, $\mathcal{M}_2^b$, $\mathcal{M}_2^d$, and $\mathcal{M}_2^e$ represent grids with refinement levels ranging from 1 to 3, while $\mathcal{M}_2^c$ is a triangular meshes with a refinement level of 1. The details of the meshes used in this test case are provided in Table 2.

3.3. Case 3: Subsurface drainage case

The third test case represents a real-world application at a larger field scale than Case 2 and is designed to assess the practicality and robustness of the VCFVM for simulating subsurface drainage systems. This scenario is based on a field experiment conducted in the Xinjiang Uygur Autonomous Region, China (86.245°E, 41.09°N), where subsurface drainage pipes were installed with different layout configurations. Three representative treatments, denoted as S_1, S_2, and S_3, are selected for numerical investigation. The location of the experimental site and the layout of the drainage system are shown in Fig. 6(a) and (b), respectively. The aquifer in this region has a thickness of 5 m, with an initial groundwater table at a depth of 2.25 m and an initial solute concentration of 3.5 g/L. During the irrigation period, the aquifer was recharged by leaching water over approximately 5 days, followed by 20 days of subsurface drainage. The cumulative volumes of drained water and solute were monitored throughout the process and used as benchmark data to validate the model performance.

For the numerical simulation, the control area of the three subsurface drainage treatments is selected as the spatial domain, defined as $\Omega = [0, 54]\,\text{m} \times [0, 85]\,\text{m}$, and the drainage duration constitutes the temporal domain, $T = [0, 20]\,\text{d}$. Groundwater flow and solute transport are simulated simultaneously, with the groundwater flow module adapted from [34]. Initial conditions are specified based on field observations, with the groundwater level set to 2.75 m and solute concentration to 3.5 g/L. No-flux boundary conditions are imposed on the outer boundaries of each treatment zone. Source terms are distributed throughout the domain to represent irrigation recharge, with distinct recharge volumes assigned to each treatment: 18.5, 17.0, and 12.3 cm, respectively. The solute concentration of the recharge water is assumed to equal the average solute concentration in the saturated soil profile, with values of 20.5, 21.3, and 33.5 g/L for the three treatments. Additionally, zones containing subsurface drainage pipes are treated as internal drainage boundaries in both the groundwater flow and solute transport modules.

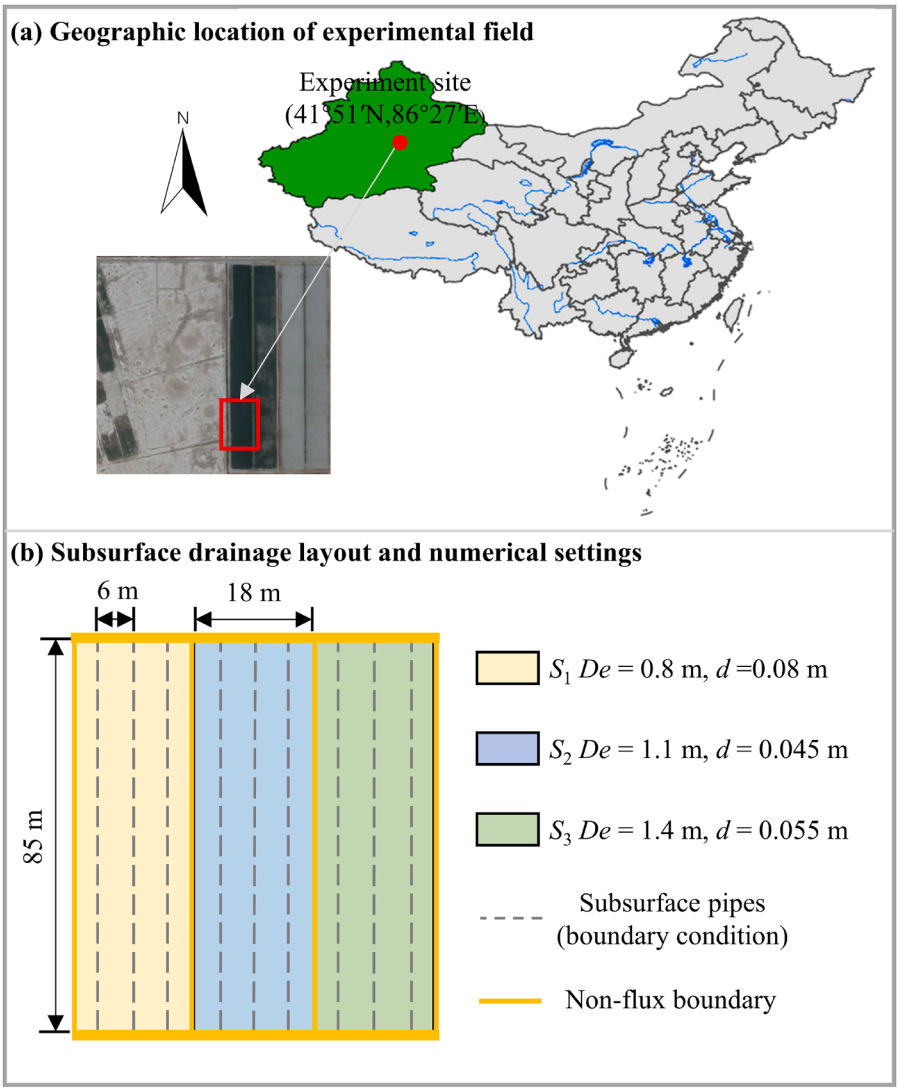
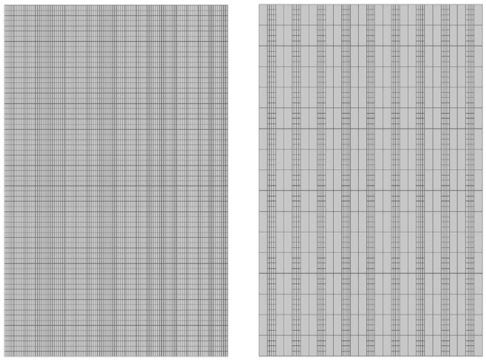


Fig. 6. Schematic diagram of Case 3: (a) geographic location of the experimental field, (b) layout of subsurface drainage and numerical settings. Note that D_e and d refer to the burial depth and diameter of the subsurface pipes, respectively.

Table 3
Values of the parameters of Case 3.

Modules	Parameters	Values
Groundwater	μ_d	0.10(-)
	$K_{xx}(K_{yy})$	0.1 m/d
Solute	θ	0.43(-)
	α_L	0.2 m
	α_T	0.02 m
Subsurface drainage	C_d	S_1 : 0.05 m/d S_2 : 0.02 m/d S_3 : 0.03 m/d

Note that μ_d is the specified yield; K_{xx} and K_{yy} are the hydraulic conductivity in xx and yy direction; θ is the soil water content; α_L and α_T are the longitudinal and transverse dispersivities; C_d is the resistance coefficient of subsurface pipe.



(a) $\mathcal{M}_3^a$ (b) $\mathcal{M}_3^b$

Fig. 7. Grids used in Case 3, including a fine grid $\mathcal{M}_3^a$ and a locally refined grid $\mathcal{M}_3^b$.

The boundary conditions described above are illustrated in Fig. 6(b), and the hydraulic and solute transport parameters used in this test case are summarized in Table 3. The case was first simulated using VCFVM on a uniform fine square grid, and the cumulative drainage volumes of both water and solute were compared against field observations for validation. To further evaluate grid flexibility, we repeated the simulation on a nonmatching grid that was locally refined around the subsurface drainage pipes, thereby highlighting the advantages of grid adaptivity near critical regions. The two types of grids used in this case, denoted as $\mathcal{M}_3^a$ (fine grid) and $\mathcal{M}_3^b$ (locally refined grid), are depicted in Fig. 7, with their key characteristics provided in Table 2.

4. Results

This section evaluates the performance of VCFVM through three representative test cases introduced earlier. Case 1 investigates the method's accuracy on nonmatching grids and compares it with other numerical schemes, highlighting the effectiveness of nonmatching grids in reducing numerical oscillations. Case 2 further examines the accuracy and computational efficiency of VCFVM across different mesh types and refinement levels. Case 3 demonstrates its applicability in a realistic, field-scale scenario.

4.1. Spatial and temporal accurate order of VCFVM

To evaluate the spatial and temporal accuracy order of the VCFVM, the simulated errors (E_1) on various grid resolution and time step size are plotted in a log-log format, and benchmarked against reference lines with slope = 1 and 2, as shown in Fig. 8.

The errors of the VCFVM and VCFVM-MLP under various scenarios are presented in Fig. 8 (a) and (b). From Fig. 8 (a) and xst(b), where the Pe ranges from 12.5 to 1.563 and from 1.25 to 0.256, it could be observed that error line of the normal VCFVM is close to reference line with a slope of 1 under high Pe, and its slope is close to 2 under low Pe, while the VCFVM with MLP consistently achieve second order accuracy under various Pe numbers. It can be concluded that the construction methods of numerical diffusive flux and advective flux in the normal VCFVM are about second order and first order accuracy, respectively. The **MLP** treatment enhances the accuracy order to second order under advection dominant scenarios.

The errors of the VCFVM discretized using Backward Euler and Crank-Nicolson are presented in Fig. 8(c) and (d). It can be observed from Fig. 8(c) and (d), where the Pe are 10 and 1, that the VCFVM with Backward-Euler consistently poses first order accurate, while the VCFVM with Crank-Nicolson achieves second order accuracy as expected.

To summarize, the normal VCFVM shows second order and first order spatial accuracy in diffusion dominated and advection dominant scenarios, respectively, and poses first order temporal accuracy. The adoption of **MLP** and Crank-Nicolson enhances the spatial and temporal accuracy order to second order, respectively.

4.2. Accuracy of VCFVM on nonmatching grids

For Case 1, Fig. 9 presents the analytical and simulated solute concentration profiles at $t = 365\,\mathrm{d}$ under various scenarios using the $\mathcal{M}_1^a$ grid, along with the corresponding simulation errors. As illustrated in Fig. 9, the solute concentration gradient increases markedly from S_1 to S_4, reflecting the rise in the Péclet number Pe. As expected, the simulation error also grows with increasing Pe.

To quantitatively assess the error, the E_1 and RE_2 were computed in the vicinity of the concentration front, specifically within the region $[0, 50]\,\mathrm{cm} \times [30, 50]\,\mathrm{cm}$, based on (54) and (55). The results, summarized in Table 4, show that although both E_1 and RE_2 increase with Pe, their maximum values-0.019 $\mathrm{g/cm^3}$ and 6.689%, respectively-remain within an acceptable range. These findings demonstrate that appropriate grid refinement near the concentration front is essential for maintaining the accuracy of the VCFVM. For this test case, a grid size of 1 cm is sufficient to produce accurate results when the diffusion coefficient satisfies $D > 0.025\,\mathrm{cm^2/d}$.

For Case 2, Fig. 10 shows the solute concentrations in the sampling profiles at $T_1 = 0.13\,\mathrm{d}$ and $T_2 = 0.53\,\mathrm{d}$, simulated using VCFVM on the locally refined mesh $\mathcal{M}_2^e$ and HYDRUS on the fine mesh $\mathcal{M}_2^a$, along with comparisons to the corresponding observed values.

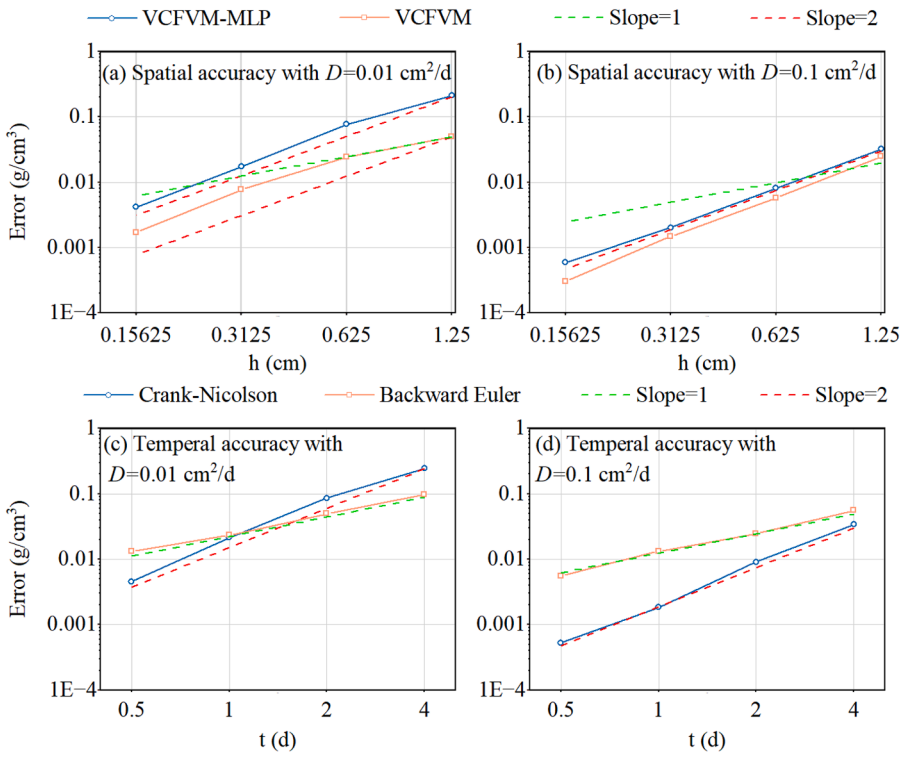


Fig. 8. Errors of the VCFVM and VCFVM with additional treatments under various spatial and temporal step sizes.

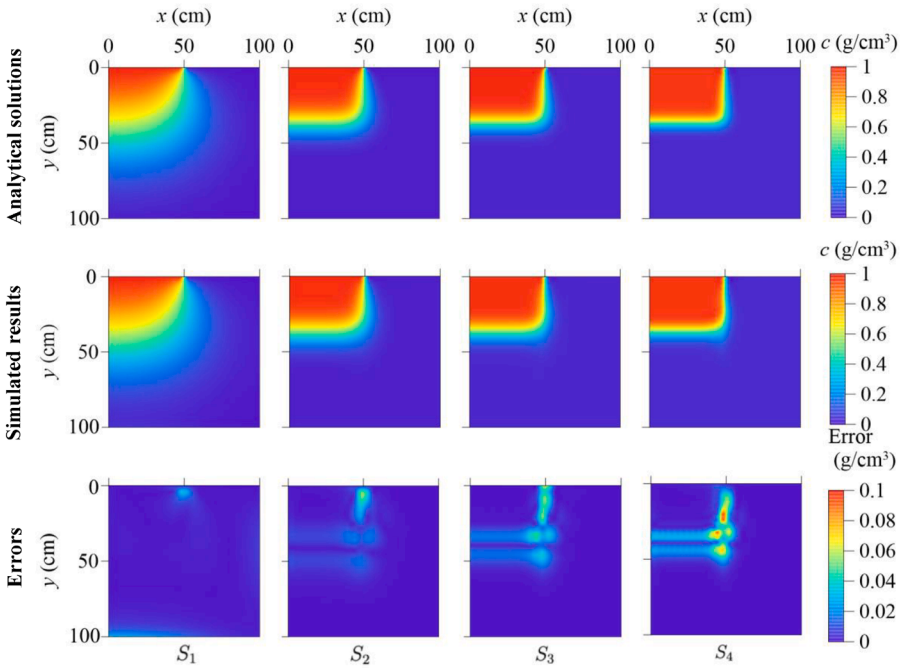


Fig. 9. Solute concentrations at $t = 365$ days for different scenarios S_1 through S_4, showing analytical solutions (first row), numerical solutions produced by the VCFVM(second row), and corresponding errors (third row).

Table 4
E_1 and RE_2 of VCFVM in the vicinity of concentration front of Case 1 using different grids under different scenarios compared to analytical solutions.

Scenarios	Grids or Methods	E_1(g/cm^3)	RE_2(%)
S_1	$\mathcal{M}_1^a$	0.001	0.162
S_2	$\mathcal{M}_1^a$	0.006	1.776
S_3	$\mathcal{M}_1^a$	0.013	4.190
S_4	$\mathcal{M}_1^a$	0.019	6.689
	$\mathcal{M}_1^b$	0.099	33.755
	$\mathcal{M}_1^b$ with **UW**	0.194	66.191
	$\mathcal{M}_1^b$ with **TL**	0.096	33.342
	$\mathcal{M}_1^b$ with **MLP**	0.168	54.210

Note that the **UW**, **TL** and **MLP** represent upstream weighting method, time step limiting method and multi-dimensional limiting process, respectively; E_1 and RE_2 represent discrete L_1-norm error and relative L_2-norm error, respectively.

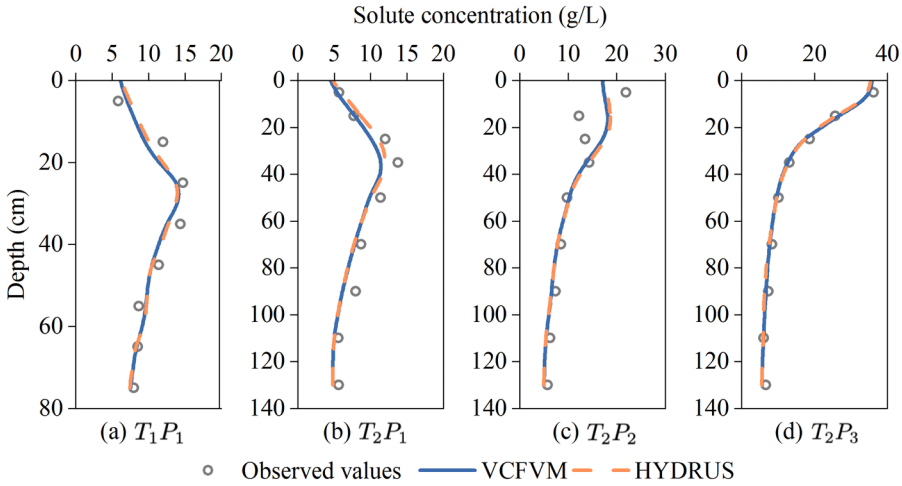


Fig. 10. The solute concentrations simulated by VCFVM on $\mathcal{M}_2^e$ and by HYDRUS on $\mathcal{M}_2^a$ compared to observed values at different time and cross sections. Noted that T_1 and T_2 represent 0.13 d and 0.53 d; P_1, P_2 and P_3 represent sampling profiles under dipper, 25 cm to dripper and 50 cm to dripper, respectively.

The results indicate that both VCFVM and HYDRUS simulations agree closely with the observations. Specifically, the average E_1 for both methods is 0.001 g/cm^3, and the average RE_2 are 11.5% for VCFVM and 11.9% for HYDRUS, demonstrating that both approaches are well-suited for this test case.

In addition, the HYDRUS results are used as a benchmark to assess the accuracy of the VCFVM on nonmatching grids. Fig. 11 presents the simulated solute concentration distributions over the entire spatial domain at different times, along with the corresponding errors, shown as heat maps. The results from VCFVM on the locally refined grid $\mathcal{M}_2^e$ and HYDRUS on the fine mesh $\mathcal{M}_2^a$ are compared. As seen in Fig. 11, the source term induces a strong solute gradient in its vicinity, and the concentration patterns simulated by VCFVM are generally consistent with those of HYDRUS. Furthermore, Fig. 11(c) and (g) reveal that the absolute error of VCFVM is relatively large near the concentration front but gradually decreases over time. Quantitatively, the E_1 at T_1 and T_2 are 0.828 and 0.766 g/L, respectively, and the corresponding RE_2 values are 4.292% and 4.371%. These results further confirm that VCFVM achieves a level of accuracy comparable to HYDRUS in this test case.

In conclusion, VCFVM demonstrates reliable accuracy on nonmatching grids in scenarios with high solute gradients, whether induced by advection-dominated transport or localized source terms, highlighting its core advantage in handling sharp fronts under complex mesh configurations.

4.3. Advantages of nonmatching grids in reducing numerical oscillations

As shown in Section 4.1, solute transport in S_4 of Case 1 is dominated by advection. This scenario was simulated on a coarse grid ($\mathcal{M}_1^b$) to evaluate the advantages of nonmatching grids in mitigating numerical oscillations.

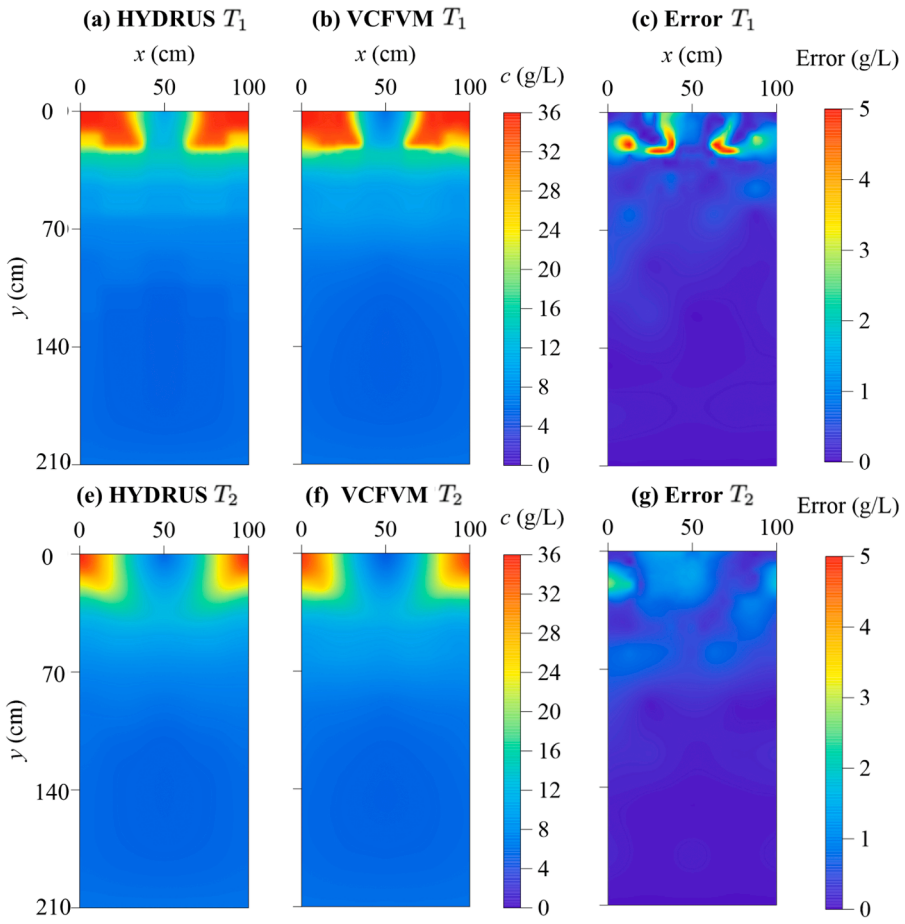


Fig. 11. Simulated solute concentrations in the spatial domain by VCFVM on $\mathcal{M}_2^c$ and HYDRUS on $\mathcal{M}_2^a$ at different times, along with the corresponding errors between VCFVM and HYDRUS. Note that T_1 and T_2 represent 0.13 d and 0.53 d, respectively.

Fig. 12 illustrates the simulated solute concentrations over 365 days using VCFVM on **CK**, **UW**, **MLP**, and **LGR**, along with the corresponding errors. Note that the simulated results obtained using the **TL** exhibited a similar pattern and error to those of the **CK**. Therefore, the detailed simulated results for the **TL** are not presented herein. Please see Secton 4 of Supplementary Material for the details. From Fig. 12(a)–(d), it is clear that the results vary significantly, with the **UW** showing notably lower solute gradients near the front compared to the other methods. The error analysis in Fig. 12(e)–(h) reveals that **LGR** has the lowest error, with an E_1 of 0.019 g/cm^3 and RE_2 of 6.689%. The **CK** and **TL** exhibit similar, higher errors, with E_1 and RE_2 around 0.1 g/cm^3 and 34%, respectively. The error of the **MLP** is higher than that of the **TL** and **CK**, whose E_1 is 0.168 g/cm^3 and RE_2 is 54.210%. The **UW** has the highest error, with an E_1 of 0.194 g/cm^3 and RE_2 of 66.191%.

To better illustrate numerical oscillation, Fig. 12(i)–(l) shows the simulated concentration along the cross-section at $x = 0$ cm. In Fig. 12(i), concentration peaks exceed the physical boundary value of 1 g/cm^3, indicating numerical oscillation. The results show that coarse grids can lead to numerical oscillations in advection-dominated flow, and refining the time step does not adequately mitigate this issue. In Fig. 12(j), (k) and (l), the peak values are effectively constrained below 1 g/cm^3, demonstrating the effectiveness of the **UW**, **MLP** and **LGR** methods in managing numerical oscillations. However, the results from both the **UW** and **MLP** are inconsistent with the analytical solution, attributed to numerical diffusion. Specifically, the higher error of the **UW** compared to the **MLP** indicates the higher degree of numerical diffusion introduced by the **UW**.

Table 2 lists the CPU times for each method. the **CK** has the shortest CPU time due to fewer vertices, while the **UW** and **MLP** requires slightly more time due to the additional calculations in Eqs. (20) or (30) at each time step. The **TL**'s CPU time is nearly double that of the **CK**, as its time step is limited to 0.5 d, compared to 1.0 d for the other methods. **LGR** has the longest CPU time, though it is still significantly faster than globally refined grids, which require 389.4 s (not shown).

In conclusion, solving advection-dominated flow scenarios on coarse grids may lead to numerical oscillations, and reducing the time step does not effectively resolve this issue while increasing CPU time. The **UW**, **MLP** and **LGR** are able to effectively prevent numerical oscillations. Among these methods, the previous two methods are more efficient but introduce additional error, which may

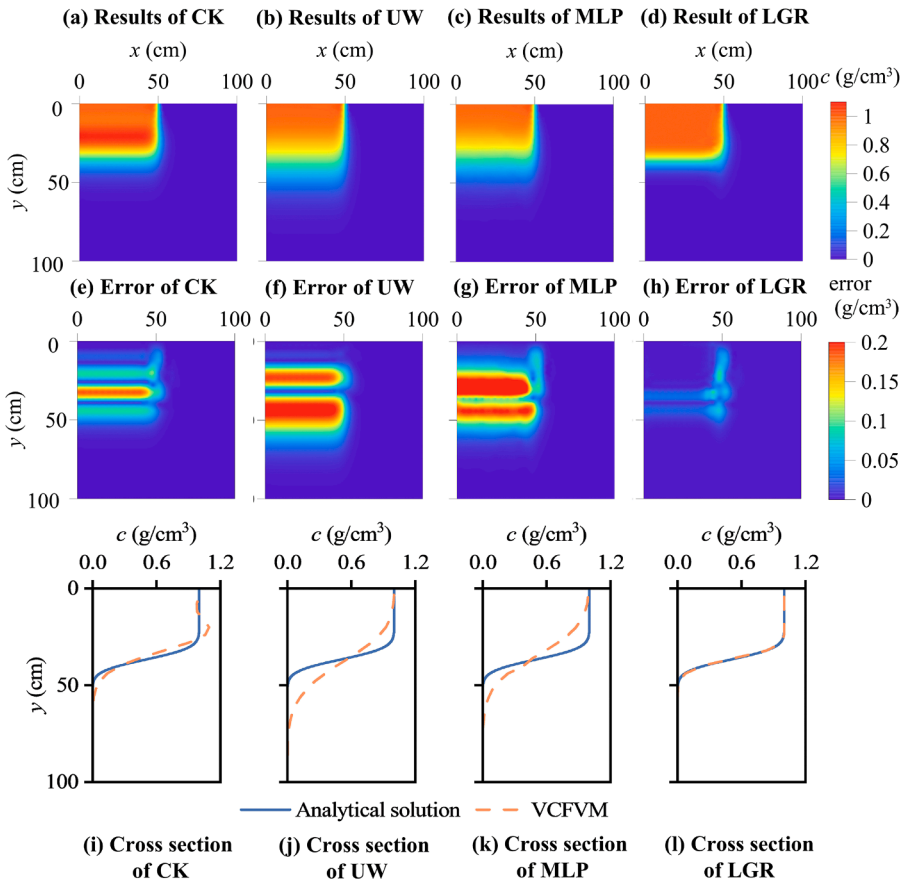


Fig. 12. Simulated solute concentrations over 365 days by the VCFVM on **CK**, **UW**, **MLP** and LGR, along with the corresponding errors in the spatial domain and cross-section. Noted that the **CK**, **UW** and **MLP** represent check, upstream weighting method, and MUSCL-type approximation approach with a Multidimensional Limiting Process, respectively.

be unacceptable in some cases, especially, the **UW**, of which the RE_2 rises to 66.191%. The **LGR**, while requiring more CPU time, do not suffer from numerical diffusion and are still much faster than globally refined grids (only 12.5% of the CPU time).

4.4. Performance of VCFVM on various nonmatching grids

The accuracy and efficiency of VCFVM were evaluated for Case 2 on nonmatching grids with varying types and refinement levels. The simulated results at different times and cross sections on these grids are compared against the benchmark results obtained on $\mathcal{M}_2^a$, as shown in Fig. 13. The $E1$ and RE_2 for each grid are calculated and summarized in Table 5, with RE_2 values plotted in Fig. 14 to show the trend of error relative to refinement level.

From both Fig. 14 and Table 5, it can be observed that the E_1 and RE_2 decrease as the refinement level increases, which underscore the importance of the grid resolution in obtaining accurate simulation results near steep gradient or small-scale processes [35,44]. Specifically, as shown in Fig. 14, for P_1, i.e., the cross section under the dripper, when the refinement level increases from 1 to 2, the RE_2 is reduced by 43% (from around 4.6% to around 2.6%). However, when the refinement level continues to increase, the accuracy does not improve significantly, indicating that for this test case, a refinement level greater than 2 does not provide significant improvement in accuracy for the P_1 cross section. On the other hand, for other locations (especially P_3), increasing the refinement level beyond 2 still reduces the RE_2. This is because only when the refinement level is high enough can the grid size at locations far from the dripper be adequately refined, as shown in Fig. 5. Additionally, the consistent RE_2 values on $\mathcal{M}_2^b$ and $\mathcal{M}_2^c$ demonstrate that the VCFVM is applicable to different types of meshes.

The CPU times for VCFVM on different meshes in Case 2 are presented in Table 2. As expected, CPU time increases with refinement level. However, even for $\mathcal{M}_2^e$ (with refinement level 3), the CPU time is only 33.4% of that for $\mathcal{M}_2^a$, demonstrating the efficiency of using nonmatching grids. To balance accuracy and efficiency, $\mathcal{M}_2^d$ is recommended for Case 2, as it achieves an RE_2 of approximately 2%, while the CPU time is only 10.5% of that for the fine mesh.

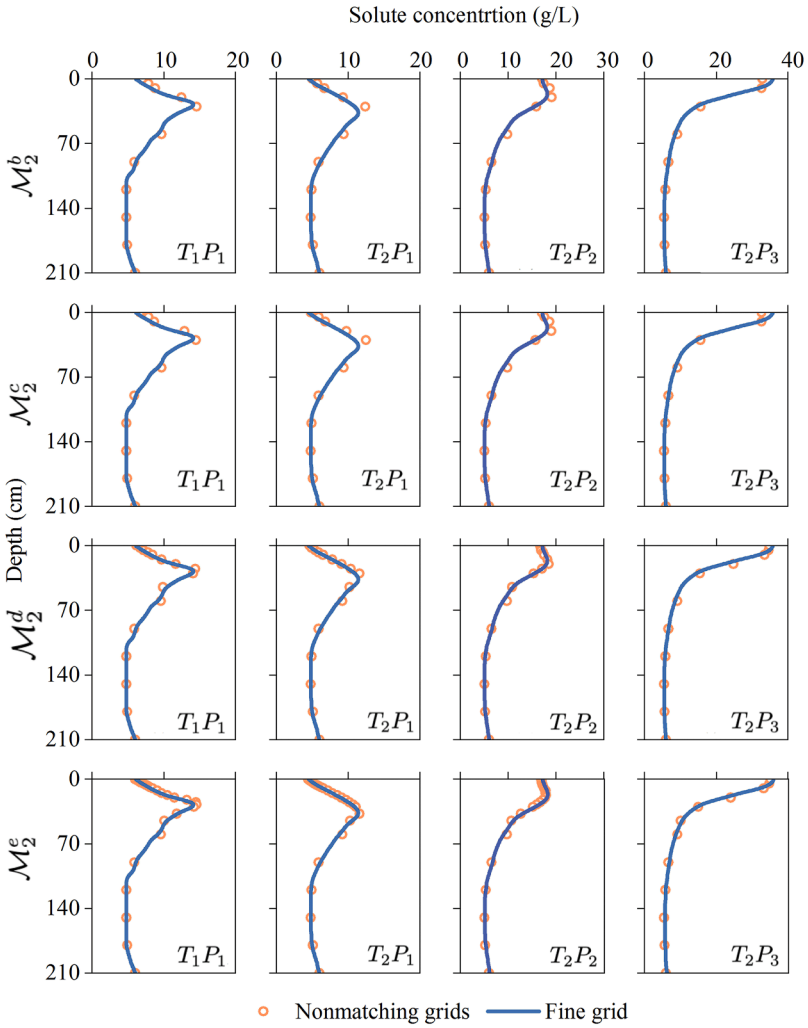


Fig. 13. Simulated solute concentrations at different times and cross sections using $\mathcal{M}_2^b, \mathcal{M}_2^c, \mathcal{M}_2^d$, and $\mathcal{M}_2^e$ compared to the values obtained with $\mathcal{M}_2^a$. T_1 and T_2 represent 0.13 d, and 0.53 d, respectively, while P_1, P_2, and P_3 correspond to the sampling profiles at the dripper, 25 cm from the dripper, and 50 cm from the dripper.

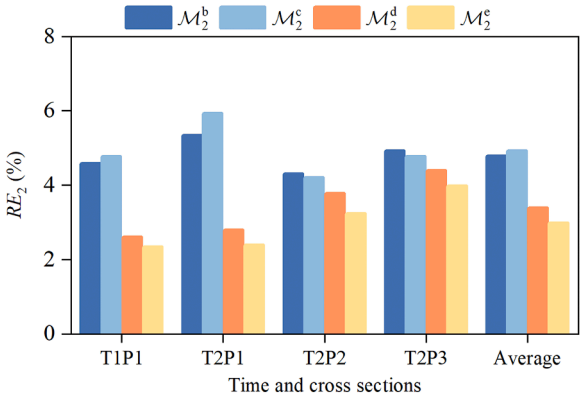


Fig. 14. The RE_2 at different time and cross sections using $\mathcal{M}_2^b$, $\mathcal{M}_2^c$, $\mathcal{M}_2^d$ and $\mathcal{M}_2^e$ compared to the corresponding values using $\mathcal{M}_2^a$. Noted that T_1 and T_2 represent 0.13 d, and 0.53 d, respectively, while P_1, P_2, and P_3 correspond to the sampling profiles at the dripper, 25 cm from the dripper, and 50 cm from the dripper; RE_2 is the discrete relative L_2-norm error.

Table 5
E_1 and RE_2 of VCFVM at different times and cross sections in Case 2 using different meshes.

Mesh	Time and cross section	E_1(g/L)	RE_2(%)
$\mathcal{M}_2^a$	T_1P_1	0.3	4.58
	T_2P_1	0.3	5.34
	T_2P_2	0.3	4.30
	T_2P_3	0.4	4.92
$\mathcal{M}_2^b$	T_1P_1	0.3	4.77
	T_2P_1	0.4	5.93
	T_2P_2	0.3	4.20
	T_2P_3	0.4	4.77
$\mathcal{M}_2^c$	T_1P_1	0.2	2.60
	T_2P_1	0.2	2.79
	T_2P_2	0.2	3.77
	T_2P_3	0.5	4.20
$\mathcal{M}_2^d$	T_1P_1	0.1	2.34
	T_2P_1	0.1	2.39
	T_2P_2	0.2	3.24
	T_2P_3	0.3	3.97

Note: $T1$ and $T2$ represent 0.13 d, and 0.53 d, respectively, while $P1$, $P2$, and $P3$ correspond to the sampling profiles at the dripper, 25 cm from the dripper, and 50 cm from the dripper; E_1 and RE_2 represent discrete L_1-norm error and relative L_2-norm error.

Table 6
MAE and RRMSE regarding to different targets simulated by VCFVM in Case 3 using different grids.

	Item	$\mathcal{M}_3^a$ vs Ob		$\mathcal{M}_3^b$ vs Ob		$\mathcal{M}_3^b$ vs $\mathcal{M}_3^a$	
		MAE	RRMSE (%)	MAE	RRMSE (%)	MAE	RRMSE (%)
Water discharge amounts	S_1	2.14 m^3	8.82	2.07 m^3	8.61	0.22 m^3	0.83
	S_2	1.72 m^3	8.25	1.98 m^3	8.78	0.28 m^3	0.90
	S_3	1.11 m^3	7.38	1.09 m^3	7.53	0.18 m^3	0.87
Solute discharge amounts	S_1	21.69 kg	7.26	26.10 kg	8.54	7.11 kg	2.24
	S_2	21.66 kg	7.72	18.45 kg	7.10	9.04 kg	2.46
	S_3	16.98 kg	7.81	21.18 kg	8.50	6.88 kg	2.25

Note: Ob is the observed vaule; MAE and RRMSE represent the mean absolute error and relative root mean square error.

In conclusion, the VCFVM performs effectively on nonmatching grids with various types and refinement levels. As refinement level increases, accuracy improves while efficiency decreases. The recommended refinement level for this test case is 2, offering an optimal balance of accuracy and efficiency.

4.5. Applicability of VCFVM in field-scale scenarios

The applicability of VCFVM to field-scale scenarios is demonstrated in Case 3 by evaluating its accuracy and computational efficiency on both the fine grid $\mathcal{M}_3^a$ and the nonmatching grid $\mathcal{M}_3^b$. To assess accuracy, the cumulative drainage volumes of water and solute computed on $\mathcal{M}_3^a$ and $\mathcal{M}_3^b$ are compared against field observations, as shown in Fig. 15. The results from both grids exhibit close agreement with the observed data and with each other.

To quantify this agreement, the relative root mean square error (RRMSE) between the simulated and observed values is reported in Table 6. The RRMSE for $\mathcal{M}_3^a$ ranges from 7.26% to 8.82%, while that for $\mathcal{M}_3^b$ ranges from 7.10% to 8.78%, confirming the reliability of both grid configurations. Furthermore, the differences between the two simulated results are evaluated using the mean absolute error (MAE) and RRMSE. As shown in Table 6, the RRMSE between $\mathcal{M}_3^a$ and $\mathcal{M}_3^b$ ranges from 0.83% to 2.46%, indicating that the use of nonmatching grids introduces minimal error. It is also observed that solute transport results are more sensitive to grid resolution than water transport results.

In terms of computational efficiency, Table 2 reports the CPU times for Case 3 across different grid configurations. As expected, CPU time increases with grid refinement. Notably, the nonmatching grid reduces computational cost by over 50% compared to the fine grid, while incurring only about 1% additional error.

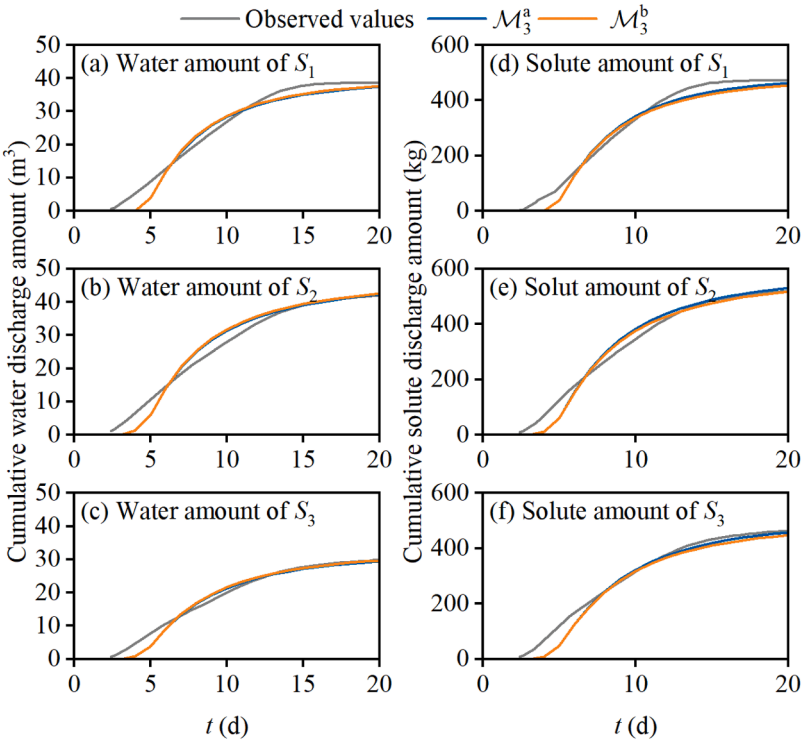


Fig. 15. Cumulative water and solute drainage amount obtained by VCFVM on $\mathcal{M}_3^a$ and $\mathcal{M}_3^b$ compared against observed values.

In summary, VCFVM proves to be both accurate and efficient for simulating solute transport in large-scale field scenarios involving localized sink terms. Additionally, the results indicate that solute transport is more grid-sensitive than water flow.

5. Discussion

Solute transport scenarios characterized by a high solute gradient are frequently encountered in practical applications. Such high gradient conditions can arise either due to high Pe or local source terms. For example, the simulation results of the Case 1 showed that the solute gradient increases with the increase of Pe (see Fig. 8). In Case 2, the solute concentration in the soil profile also exhibited a pronounced solute gradient (see Fig. 10), which demonstrated that source term, even water source term with negligible solute concentration, could induce high solute gradients.

In those scenarios involving local source terms, grid refinement in the vicinity of these source terms is necessary to ensure high accuracy [34,44]. The simulation results of Case 2 on various nonmatching grids demonstrated that the accuracy improves with the increase of refinement level, as illustrated in Fig. 13. However, the regions required refinement, i.e., the sizes of the source terms, generally occupy only a minor portion of the simulation domain. Taking Case 2 and Case 3, which represent site scale and field scale, for instance, the dimensions of the dripper and the subsurface pipes are 1-2 orders of magnitude smaller than those of the entire simulation domain. This disparity could be even more pronounced at watershed scale simulations [12]. In this context, implementing grid refinement through nonmatching grids could circumvent transition between fine and coarse grids, therefore enhance computational efficiency [11]. Case 2 and Case 3 showed that, the proposed model could save about 90% and 50% CPU time on nonmatching grids compared to fine grids, respectively. Such grids have also been adopted in certain previous models or algorithms. However, in contrast to those models that set additional ghost nodes [45] or conduct additional iterations [46] to ascertain the unknown variables at the nonmatching interface, the algorithm proposed here is compatible with arbitrary polygonal grids, including nonmatching grids, without necessitating any supplementary operation at the nonmatching interface. This characteristic is proven to have advantages in relieving errors arising from the additional treatment [34,35].

In scenarios dominated by advection, high resolution grids near the regions characterized by sharp concentration transitions is also essential to guarantee accuracy [47]. For example, in Case 1, the simulation error on coarse grid is approximately 34% when Pe reached 16, and the error primarily concentrated in the solute front region.Using nonmatching grids near the concentration front, which helps to restrict the Pe lower than 4, could limit the relative error below 7%. This finding aligns closely with some research which reported that maintaining Pe at a value not exceeding 5 is crucial to keep accuracy in advection dominated scenarios [5,48]. Additionally, with the context of advection-dominated scenarios, apart from simulation error, numerical oscillation is another issue that garnered considerable attention from researchers [49]. Besides LGR, UW and TL are two widely-used ways to reduce numerical oscillation. In Case 1 of this study, we compared the performance of TL and conventional UW in reducing numerical oscillation against

LGR. Results revealed that TL does not adequately mitigate this issue, and both LGR and UW are capable of restricting the peak value of the solute concentration below that on soil surface, indicating the effectiveness of LGR and UW in addressing this issue. However, results of Case 1 also showed that the error of UW significant increased in a region larger than the solute front (see Fig. 11). This is attributed to numerical diffusion [50,51]. According to some previous studies, the phenomenon of numerical diffusion is subject to the influence of the numerical algorithm employed [52]. Regarding UW, though effectively suppressing numerical oscillation, its application in high Pe requires caution due to the potential for excessive numerical diffusion [53]. Our study demonstrated that in advection dominated solute transport scenarios with Pe over 16, the additional error introduced by UW is non-negligible. Therefore, LGR emerges as a good alternative to reduce numerical oscillation without inducing additional error associated with numerical diffusion.

Based on the discussion above, the algorithm proposed in this study offers a good alternative for simulating solute transport involving high solute gradient on nonmatching grid, demonstrating advantages in terms of ensuring accuracy, improving efficiency, and mitigating numerical oscillation.

6. Conclusion

In this study, we developed a numerical algorithm based on the vertex-centered finite volume method (VCFVM), designed for simulating solute transport in porous media on arbitrary polygonal meshes. The accuracy and efficiency of the proposed method were systematically evaluated through three representative test cases characterized by sharp solute gradients. The key conclusions are as follows:

1. The VCFVM algorithm accurately simulates solute transport problems with steep concentration gradients using nonmatching grids. The normal VCFVM with Backward Euler method shows second and first spatial accuracy order in diffusion dominated and in advection dominant scenarios, respectively, and poses first order temporal accuracy. Adopting Multidimensional Limiting Process (**MLP**) and Crank-Nicolson enhances the spatial and temporal accuracy order to second order.
2. For advection-dominated solute transport, VCFVM effectively mitigates numerical oscillations by maintaining the Péclet number below 4 via nonmatching grid strategies. Compared to globally refined grids, this approach reduces computational time to just 14.5%. Furthermore, unlike conventional upstream schemes or slope limiter method, it avoids additional errors from numerical diffusion, thereby enhancing accuracy.
3. The accuracy of VCFVM improves with increased refinement of the nonmatching grid, albeit with reduced efficiency. The method also demonstrates consistent performance across meshes refined with cells of different shapes.
4. The VCFVM is well-suited for large-scale field applications involving localized source/sink terms. In Case 3, it achieved a ∼50% reduction in CPU time with only about 1% additional error, highlighting its excellent balance between accuracy and computational efficiency.

Moreover, the algorithm can be tightly integrated with previously developed groundwater and soil water flow models, showing strong potential for practical applications involving sharp gradients in both flow and solute transport. However, this study primarily focus on the construction of the numerical diffusive flux for solute transport and its coupling with groundwater and soil water flow. The integration of the VCFVM with more numerical technics for handling advective flux and temporal integration, such as Riemann solvers, Runge-Kutta time integration and so on, remains to be further studied.

CRediT authorship contribution statement

Yingzhi Qian: Writing – original draft, Validation, Funding acquisition; **Xiaoping Zhang:** Writing – review & editing, Methodology, Funding acquisition; **Yan Zhu:** Writing – review & editing, Supervision, Funding acquisition, Data curation; **Lili Ju:** Methodology, Funding acquisition, Conceptualization; **Jiesheng Huang:** Supervision, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

Y. Qian's research is partially supported by National Natural Science Foundation of China through 52409066. X. Zhang's research is partially supported by Natural Science Foundation of China through 12571467 and Hubei Provincial Natural Science Foundation of China through Grant 2024AFB749. Y. Zhu's research is partially supported by Natural Science Foundation of China through 52422903.

References

[1] M. Qadir, E. Quillérou, V. Nangia, Economics of salt-induced land degradation and restoration, Nat. Resour. Forum 38 (4) (2014) 282–295. https://doi.org/10.1111/1477-8947.12054

[2] K. Mmolawa, D. Or, Root zone solute dynamics under drip irrigation: a review, Plant Soil. 222 (1-2) (2000) 163–190. https://doi.org/10.1023/A:1004756832038

[3] S. Altinok, E. Yurtseven, S. Avci, The effects of different irrigation water salinities and leaching ratios on green and dry forage yields of alfalfa (medicago sativa l.), J. Agric. Forestry 61 (1) (2015) 85–90. https://doi.org/10.17707/AgricultForest.61.1.10.

[4] H.P. Ritzema, Drainage Principles and Applications, International Institute for Land Reclamation and Improvement, Netherlands, Netherlands, 2006. https://edepot.wur.nl/75186.

[5] J. Simunek, M. van Genuchten, M. Sejna, The HYDRUS software package for simulating the two- and three-dimensional movement of water, heat, and multiple solutes in variably-saturated media, Proc. 11th Int. Meshing Roundtable 44 (6) (2006) 797–802. https://doi.org/10.1111/j.1745-6584.2005.00179.x

[6] C. Zheng, Recent developments and future directions for MT3DMS and related transport codes, Groundwater 47 (2009) 620–625. https://doi.org/10.1111/j.1745-6584.2009.00602.x

[7] I. Kramer, Y. Mau, Soil degradation risks assessed by the SOTE model for salinity and sodicity, Water Resour. Res. 56 (10) (2020) e2020WR027456. https://doi.org/10.1029/2020WR027456

[8] Y. Liu, C. Ao, W. Zeng, A. Kumar Srivastava, T. Gaiser, J. Wu, J. Huang, Simulating water and salt transport in subsurface pipe drainage systems with HYDRUS-2D, J. Hydrol. 592 (2021) 125823. https://doi.org/10.1016/j.jhydrol.2020.125823

[9] Y. Yang, Y. Zhu, J. Wu, Development and application of a new package for MODFLOW-LGR-MT 3D for simulating regional groundwater and salt dynamics with subsurface drainage systems, Agric. Water Manage. 260 (2022) 107330. https://doi.org/10.1016/j.agwat.2021.107330

[10] J. Simunek, M. van Genuchten, M. Sejna, Recent developments and applications of the HYDRUS computer software packages, Vadose Zone J. 15 (7) (2016) 1–25. https://doi.org/10.2136/vzj2016.04.0033

[11] M.A. Sbai, N. Amraoui, Automatic generation of locally refined composite grids for efficient solute transport modeling, Groundwater 58 (1) (2020) 35–45. https://doi.org/10.1111/gwat.12882

[12] X. Yang, H. Sun, Y. Yang, Recent progress in multi-scale modeling and simulation of flow and solute transport in porous media, Wiley Interdiscip. Rev.-Water 8 (2021). https://doi.org/10.1002/wat2.1561

[13] X. Zhang, F. Ma, S. Yin, Application of upscaling methods for fluid flow and mass transport in multi-scale heterogeneous media: a critical review, Appl. Energy 303 (4) (2021) 117603. https://doi.org/10.1016/j.apenergy.2021.117603

[14] F.R.L. Contreras, U.A.O. Vaz, G.L.S.S. Pacheco, A.R.E. Antunes, P.R.M. Lyra, D.K.E. Carvalho, A high-resolution multidimensional finite volume scheme coupled to a nonlinear two-point flux approximation method for the numerical simulation of groundwater contaminant transport using unstructured 2D meshes, Adv. Water Resour. 181 (2023) 104559. https://doi.org/10.1016/j.advwatres.2023.104559

[15] H.J. Diersch, FEFLOW - Finite Element Modeling of Flow, Mass and Heat Transport in Porous and Fractured Media, Springer, 2014. https://doi.org/10.1007/978-3-642-38739-5

[16] P. Brunner, C.T. Simmons, HydroGeoSphere: a fully integrated, physically based hydrological model, Groundwater 50 (2) (2012) 170–176. https://doi.org/10.1111/j.1745-6584.2011.00882.x

[17] S. Panday, C.D. Langevin, R.G. Niswonger, M. Ibaraki, J.D. Hughes, MODFLOW-USG Version 1: An Unstructured Grid Version of MODFLOW for Simulating Groundwater Flow and Tightly Coupled Processes Using a Control Volume Finite-Difference Formulation, U.S. Geological Survey Techniques and Methods, 2013. https://pubs.usgs.gov/tm/06/a45.

[18] W. Feng, T.L. Lewis, S.M. Wise, Discontinuous Galerkin derivative operators with applications to second-order elliptic problems and stability, Math. Meth. Appl. Sci. 38 (2015) 5160–5182. https://doi.org/10.1002/mma.3440

[19] F. Székely, Numerical modeling of drainage schemes in layered formations: a Kuwait city case study, Groundwater 37 (1999) 879–883. https://doi.org/10.1111/j.1745-6584.1999.tb01186.x

[20] S. Mehl, M.C. Hill, Development and evaluation of a local grid refinement method for block-centered finite-difference groundwater models using shared nodes, Adv. Water Resour. 25 (2002) 497–511. https://doi.org/10.1016/S0309-1708(02)00021-0

[21] Y. Cao, M. Gunzburger, X. He, X. Wang, Parallel, non-iterative, multi-physics domain decomposition methods for time dependent Stokes-Darcy systems, Math. Comput. 83 (288) (2014) 1617–1644. https://doi.org/10.1090/S0025-5718-2014-02779-8

[22] M.T. Balhoff, S.G. Thomas, M.F. Wheeler, Mortar coupling and upscaling of pore-scale models, Comput. Geosci. 12 (1) (2008) 15–27. https://doi.org/10.1007/s10596-007-9058-6

[23] Y. Mehmani, M.T. Balhoff, Bridging from pore to continuum: a hybrid mortar domain decomposition framework for subsurface flow and transport, Multiscale Model. Simul. 12 (2) (2014) 667–693. https://doi.org/10.1137/13092424X

[24] T. Hou, X. Wu, A multiscale finite element method for elliptic problems in composite materials and porous media, J. Comput. Phys. 134 (1) (1997) 169–189. https://doi.org/10.1006/jcph.1997.5682

[25] P. Jenny, S.H. Lee, H.A. Tchelepi, Multi-scale finite-volume method for elliptic problems in subsurface flow simulation, J. Comput. Phys. 187 (1) (2003) 47–67. https://doi.org/10.1016/S0021-9991(03)00075-5

[26] S. Fu, E. Chung, L. Zhao, Generalized multiscale finite element method for highly heterogeneous compressible flow, Multiscale Model. Simul. 20 (4) (2022) 1437–1467. https://doi.org/10.1137/21M1438475.

[27] C. Luo, Y. Shi, D. Timlin, A multiscale finite element method for coupled heat and water transfer in heterogeneous soils, J. Hydrol. 612 (B) (2022) 128028. https://doi.org/10.1016/j.jhydrol.2022.128028

[28] D. Spiridonov, M. Vasilyeva, M. Wang, Mixed generalized multiscale finite element method for flow problem in thin domains, J. Comput. Appl. Math. 416 (2022) 114577. https://doi.org/10.1016/j.cam.2022.114577

[29] L.A. Poveda, J. Galvis, E. Chung, A second-order exponential integration constraint energy minimizing generalized multiscale method for parabolic problems, J. Comput. Phys. 502 (2024) 112796. https://doi.org/10.1016/j.jcp.2024.112796

[30] M. Mosharaf-Dehkordi, A fixed point multi-scale finite volume method: application to two-phase incompressible fluid flow through highly heterogeneous porous media, J. Comput. Phys. 462 (2022) 111219. https://doi.org/10.1016/j.jcp.2022.111219

[31] Z. Mehrdoost, Multiscale simulation of highly heterogeneous subsurface flow with physically adaptive 3D unstructured grids, Adv. Water Resour. 184 (2024) 104611. https://doi.org/10.1016/S0167-5648(04)80130-8

[32] O. Chaabi, M.A. Kobaisi, Algorithmic monotone multiscale finite volume methods for porous media flow, J. Comput. Phys. 499 (2024) 112739. https://doi.org/10.1016/j.jcp.2023.112739

[33] Y. Zha, J. Yang, J. Zeng, C.-H.M. Tso, W. Zeng, L. Shi, Review of numerical solution of Richardson-Richards equation for variably saturated flow in soils, Water 6 (5) (2019) e1364. https://doi.org/10.1002/wat2.1364

[34] Y. Qian, Y. Zhu, X. Zhang, J. Wu, M. Ye, W. Mao, J. Wu, J. Huang, J. Yang, A local grid-refined numerical groundwater model based on the vertex-centred finite-volume method, Adv. Water Resour. 173 (2023) 104392. https://doi.org/10.1016/j.advwatres.2023.104392

[35] Y. Qian, X. Zhang, Y. Zhu, L. Ju, A. Guadagnini, J. Huang, A novel vertex-centered finite volume method for solving Richards' equation and its adaptation to local mesh refinement, J. Comput. Phys. 501 (2024) 112766. https://doi.org/10.1016/j.jcp.2024.112766

[36] M.T. van Genuchten, A closed-form equation for predicting the hydraulic conductivity of unsaturated soils, Soil Sci. Soc. Am. J. 44 (5) (1980) 892–898. https://doi.org/10.2136/sssaj1980.03615995004400050002x

[37] N. Sun, W. Yeh, A proposed upstream weight numerical method for simulating pollutant transport in groundwater, Water Resour. Res. 19 (6) (1983). https://doi.org/10.1029/WR019i006p01489

[38] J. Noorishad, C. Tsang, F.P. Perrochet, A. Musy, A perspective on the numerical solution of convection-dominated transport problems: a price to pay for the easy way out, Water Resour. Res. 28 (2) (1992) 551–561. https://doi.org/10.1029/91WR02730

[39] P. Perrochet, D. BéRod, Stability of the standard Crank-Nicolson-Galerkin scheme applied to the diffusion-convection equation: some new insights, Water Resour. Res. 29 (9) (1993) 3291–3297. .
[40] J.S. Park, S.H. Yoon, C. Kim, Multi-dimensional limiting process for hyperbolic conservation laws on unstructured grids, J. Comput. Phys. 229 (3) (2010) 788–812. https://doi.org/10.1016/j.jcp.2009.10.011
[41] A. Shokri, W.E. Bardsleya, Development, testing and application of DrainFlow: a fully distributed integrated surface-subsurface flow model for drainage study, Adv. Water Resour. 96 (2016) 299–315. https://doi.org/10.1016/j.advwatres.2016.04.013
[42] X. Zhang, S. Su, J. Wu, A vertex-centered and positivity-preserving scheme for anisotropic diffusion problems on arbitrary polygonal grids, J. Comput. Phys. 344 (2017) 419–436. https://doi.org/10.1016/j.jcp.2017.04.070
[43] I. Javandel, C. Doughty, C. Tsang, Groundwater Transport: Handbook of Mathematical Models, Am. Geophys. Union, Washington, D.C., Washington, D.C., 1984. https://doi.org/10.1029/WM010
[44] C. Foucart, A. Charous, P.F.J. Lermusiaux, Deep reinforcement learning for adaptive mesh refinement, J. Comput. Phys. 491 (2023) 112381. https://doi.org/10.1016/j.jcp.2023.112381
[45] S. Mehl, M.C. Hill, S.A. Leake, Comparison of local grid refinement methods for MODFLOW, Groundwater 44 (6) (2006) 792–796. https://doi.org/10.1111/j.1745-6584.2006.00192.x
[46] G. Kourakos, T. Harter, Simulation of unconfined aquifer flow based on parallel adaptive mesh refinement, Water Resour. Res. 57 (12) (2021) e2020WR029354. https://doi.org/10.1029/2020WR029354
[47] L. Pan, A.W. Warrick, P.J. Wierenga, Finite element methods for modeling water flow in variably saturated porous media: numerical oscillation and mass-distributed schemes, Water Resour. Res. 32 (6) (1996) 1883–1889. https://doi.org/10.1029/96WR00753
[48] J.D. Hunt, S.Z. Lu, Numerical modeling of cellular/dendritic array growth: spacing and structure predictions, Metall. Mater. Trans. A-Phys. Metall. Mater. Sci. 27 (1996) 611–623. https://doi.org/10.1007/BF02648950
[49] F.R.L. Contreras, D.K.E. Carvalho, G. Galindez-Ramirez, P.R.M. Lyra, A non-linear finite volume method coupled with a modified higher order MUSCL-type method for the numerical simulation of two-phase flows in non-homogeneous and non-isotropic oil reservoirs, Comput. Math. Appl. 92 (2021) 120–133. https://doi.org/10.1016/j.camwa.2021.03.023
[50] D.E. Apushkinskaya, G.G. Lazareva, V.A. Okishev, Influence of numerical diffusion on the growth rate of viscous fingers in the numerical implementation of the peaceman model by the finite volume method, J. Math. Sci. 283 (2024) 1–9. https://doi.org/10.1007/s10958-024-07235-5
[51] A. Khrabry, I.D. Kaganovich, S. Raman, E. Turkoz, D. Graves, Numerical method for modeling nucleation and growth of particles that prevents numerical diffusion, [Preprint] (2024). https://doi.org/10.48550/arXiv.2401.05559
[52] K. Kotrasova, V. Michalcova, The study of the numerical diffusion in computational calculation, MATEC Web Conf. 310 (2020) 00039. https://doi.org/10.1051/matecconf/202031000039
[53] T.H. Tsang, A. Rao, Comparison of different numerical schemes for condensational growth of aerosols, Aerosol Sci. Technol. 9 (3) (1988) 271–277. https://doi.org/10.1080/02786828808959214